

Computer Science

Chapter 5- The internet and its uses

By: Mr. Nouredine Tadjerout

Chapter 5- The internet and its uses

5.1 The internet and the world wide web

5.2 Digital currency

5.3 Cyber security

Learning Objectives

5 The internet and its uses

5.1 The internet and the world wide web

Candidates should be able to:

- 1 Understand the difference between the internet and the world wide web
- 2 Understand what is meant by a uniform resource locator (URL)
- 3 Describe the purpose and operation of hypertext transfer protocol (HTTP) and hypertext transfer protocol secure (HTTPS)
- 4 Explain the purpose and functions of a web browser

Notes and guidance

- The internet is the infrastructure
- The world wide web is the collection of websites and web pages accessed using the internet
- A URL is a text-based address for a web page; it can contain the protocol, the domain name and the web page/file name
- The main purpose of a web browser is to render hypertext markup language (HTML) and display web pages
- Functions include:
 - storing bookmarks and favourites
 - recording user history
 - allowing use of multiple tabs
 - storing cookies
 - providing navigation tools
 - providing an address bar

Learning Objectives

5.1 The internet and the world wide web continued

Candidates should be able to:

- 5 Describe how web pages are located, retrieved and displayed on a device when a user enters a URL
- 6 Explain what is meant by cookies and how they are used, including session cookies and persistent cookies

Notes and guidance

- Including the role of:
 - the web browser
 - IP addresses
 - domain name server (DNS)
 - web server
 - HTML
- Cookies are used for functions, including:
 - saving personal details
 - tracking user preferences
 - holding items in an online shopping cart
 - storing login details

Learning Objectives

5.2 Digital currency

Candidates should be able to:

- 1 Understand the concept of a digital currency and how digital currencies are used
- 2 Understand the process of blockchain and how it is used to track digital currency transactions

Notes and guidance

- A digital currency is one that only exists electronically
- Blockchain, in its basic form, is a digital ledger, that is a time-stamped series of records that cannot be altered

Learning Objectives

5.3 Cyber security

Candidates should be able to:

- 1 Describe the processes involved in, and the aim of carrying out, a range of cyber security threats

Notes and guidance

- Including:
 - brute-force attack
 - data interception
 - distributed denial of service (DDoS) attack
 - hacking
 - malware (virus, worm, Trojan horse, spyware, adware, ransomware)
 - pharming
 - phishing
 - social engineering

Learning Objectives

5.3 Cyber security continued

Candidates should be able to:

- 2 Explain how a range of solutions are used to help keep data safe from security threats

Notes and guidance

- Including:
 - access levels
 - anti-malware including anti-virus and anti-spyware
 - authentication (username and password, biometrics, two-step verification)
 - automating software updates
 - checking the spelling and tone of communications
 - checking the URL attached to a link
 - firewalls
 - privacy settings
 - proxy-servers
 - secure socket layer (SSL) security protocol

5.1 The internet and the world wide web

Learning Objectives

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5.1 The internet and the world wide web

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5.1 The internet and the world wide web

Differences between the internet and the World Wide Web:

Internet

The Internet, sometimes called simply "the Net," is a worldwide system of computer networks -- a network of networks in which users at any one computer can, if they have permission, get information from any other computer (and sometimes talk directly to users at other computers. **Think of the internet as the roads that connect towns and cities together.**

<https://www.youtube.com/watch?v=Dxcc6ycZ73M>



5.1 The internet and the world wide web

Differences between the internet and the World Wide Web:

World Wide Web (WWW)

The World Wide Web, commonly known as the Web, is an information system enabling documents and other web resources to be accessed over the Internet. Documents and downloadable media are made available to the network through web servers and can be accessed by programs such as web browsers. **The world wide web contains the things you see on the roads like houses and shops.**

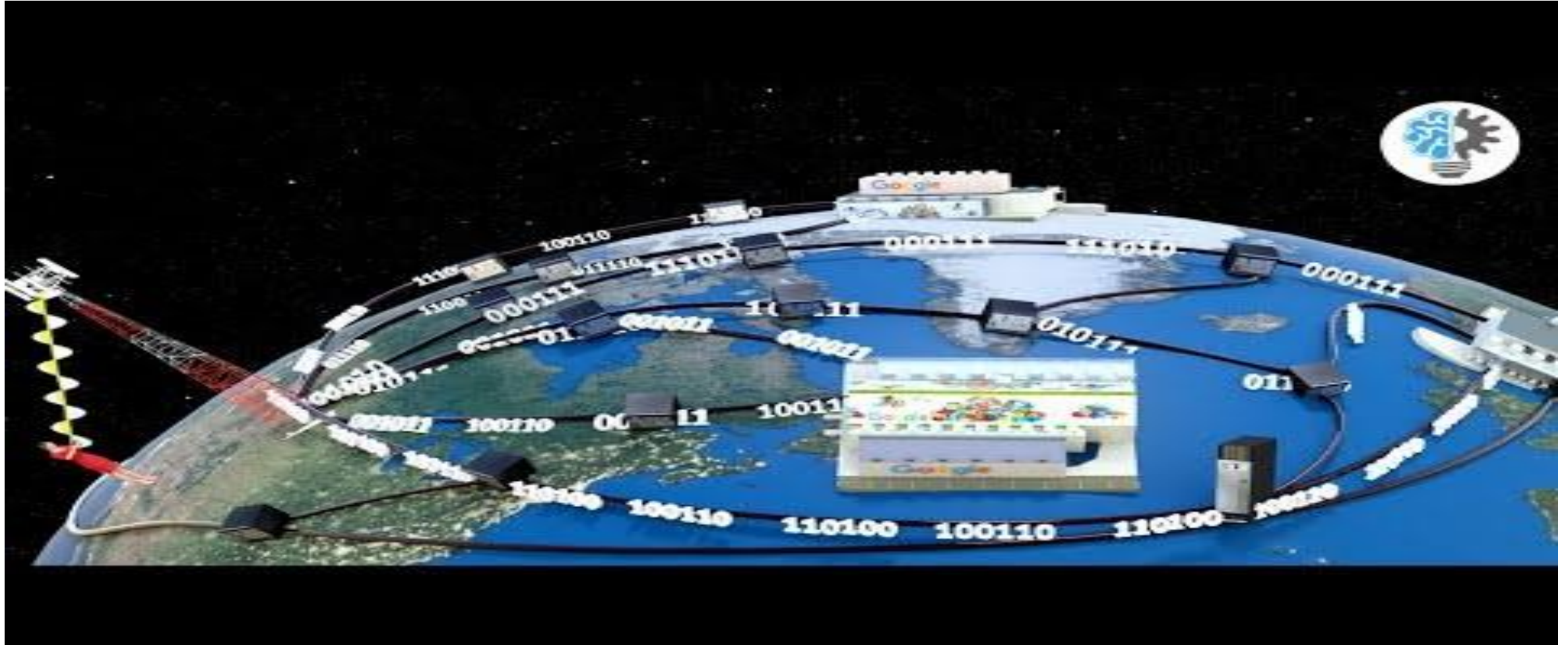
<https://www.youtube.com/watch?v=J8hzJxb0rpc>



5.1 The internet and the world wide web

How does the INTERNET work?

<https://youtu.be/x3c1ih2NJEg>



5.1 The internet and the world wide web

Differences between the internet and the World Wide Web:

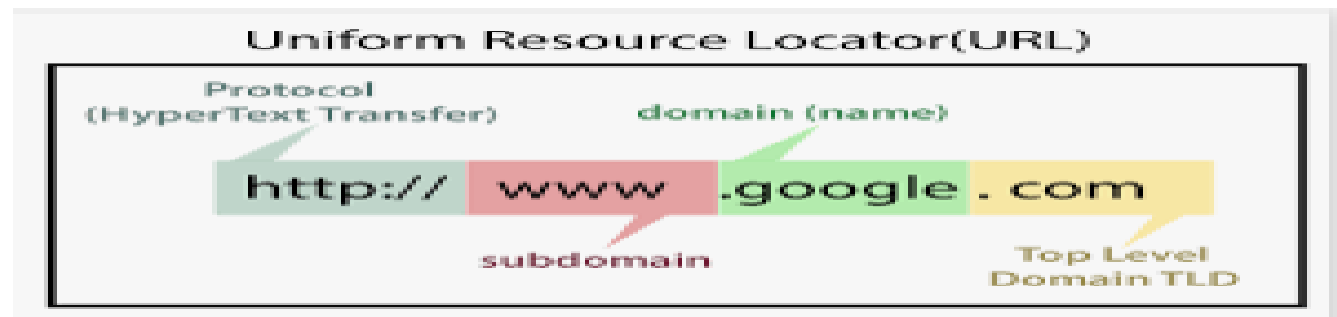
World Wide Web (WWW)

The **World Wide Web** (WWW or 'web' for short) is the part of the internet that you can access using a **web browser** such as **Internet Explorer** or **Firefox**. It consists of a large number of web servers that host websites. Each website will normally consist of a number of web pages. A web page can contain text, images, video, animation and sound.

Accessing web pages:

You can access a website or web page by typing its URL (Uniform Resource Locator) into the address bar of your browser. An example of a URL is **https://www.epsom.com/**.

URLs will have the format '**http**' and a **domain** (such as **.com**).



5.1 The internet and the world wide web

Differences between the internet and the World Wide Web:

INTERNET	World Wide Web
Internet is a global network of networks	WWW stands for World wide Web
Internet is a means of connecting a computer to any other computer anywhere in the world.	World Wide Web which is a collection of information which is accessed via the Internet
Internet is infrastructure.	WWW is service on top of that infrastructure
Internet can be viewed as a big book-store	Web can be viewed as collection of books on that store
At some advanced level, to understand we can think of the Internet as hardware.	At some advanced level, to understand we can think of the WWW as software
Internet is primarily hardware-based.	WWW is more software-oriented as compared to the Internet.
It is originated sometimes in late 1960s.	English scientist Tim Berners-Lee invented the World Wide Web in 1989
Internet is superset of WWW.	WWW is a subset of the Internet
The first version of the Internet was known as ARPANET.	In the beginning WWW was known as NSFNET.
Internet uses IP address	WWW uses HTTP

5.1 The internet and the world wide web

Uniform resource locators (URLs)

A URL (Uniform Resource Locator) is a **unique identifier used to locate a resource on the Internet**. It is also referred to as a web address. URLs consist of multiple parts -- including a protocol and domain name -- that tell a web browser how and where to retrieve a resource.

https://www.youtube.com/watch?v=5Jr-_Za5yQM



Understanding
URLS

← → ↻ https://|

5.1 The internet and the world wide web

HTTP and HTTPS

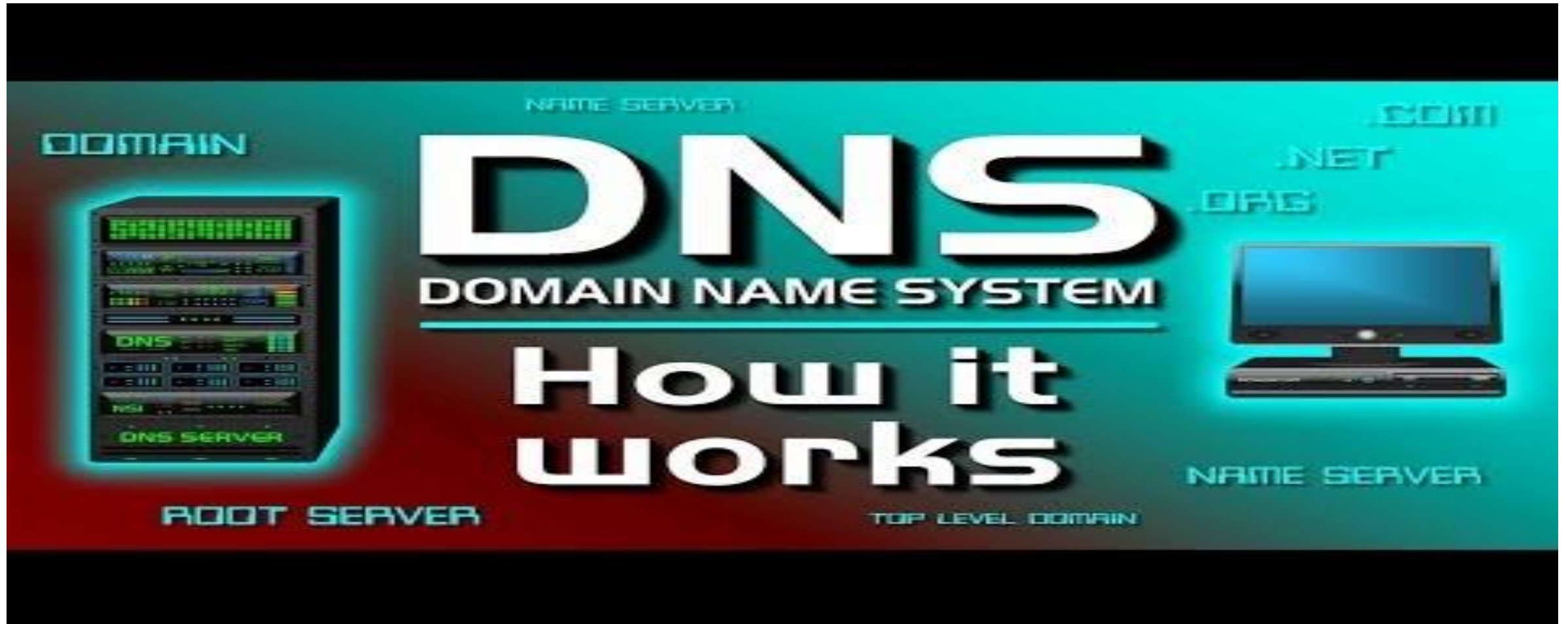
<https://www.youtube.com/watch?v=hExRDVZHhig>



5.1 The internet and the world wide web

DNS (Domain Name System)

<https://www.youtube.com/watch?v=mpQZVYPuDGU>



5.1 The internet and the world wide web

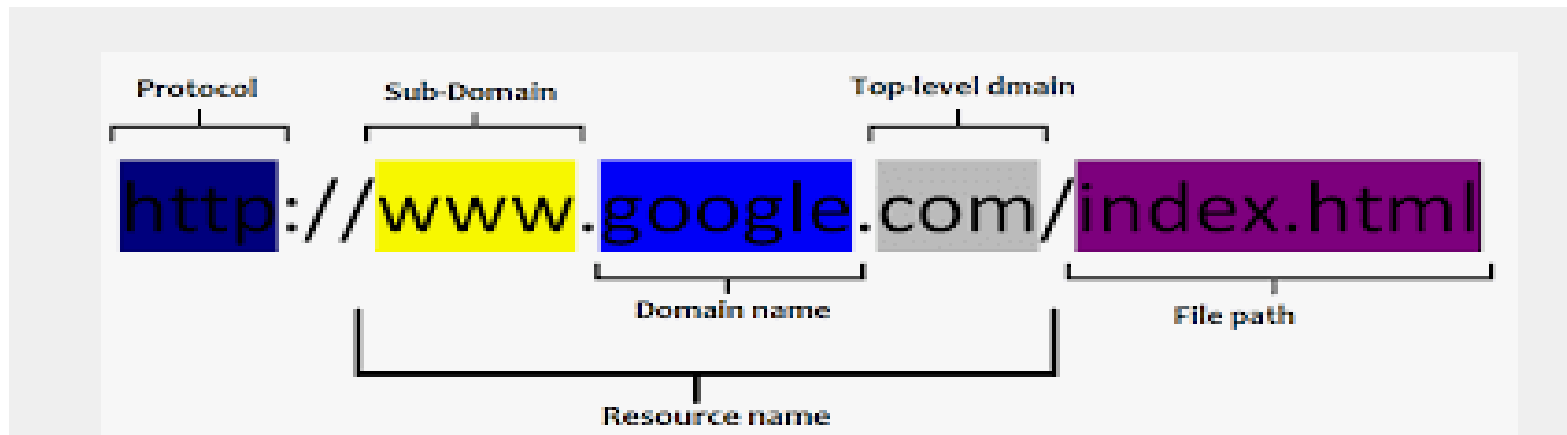
Uniform resource locators (URLs)

protocol://**website address**/**path**/**file name**

The protocol is usually either **http** or **https**.

The website address is:

- domain host (www),
- domain name (website name),
- domain type (**.com**, **.org**, **.net**, **.gov**, ...etc), example country code: my, uk)



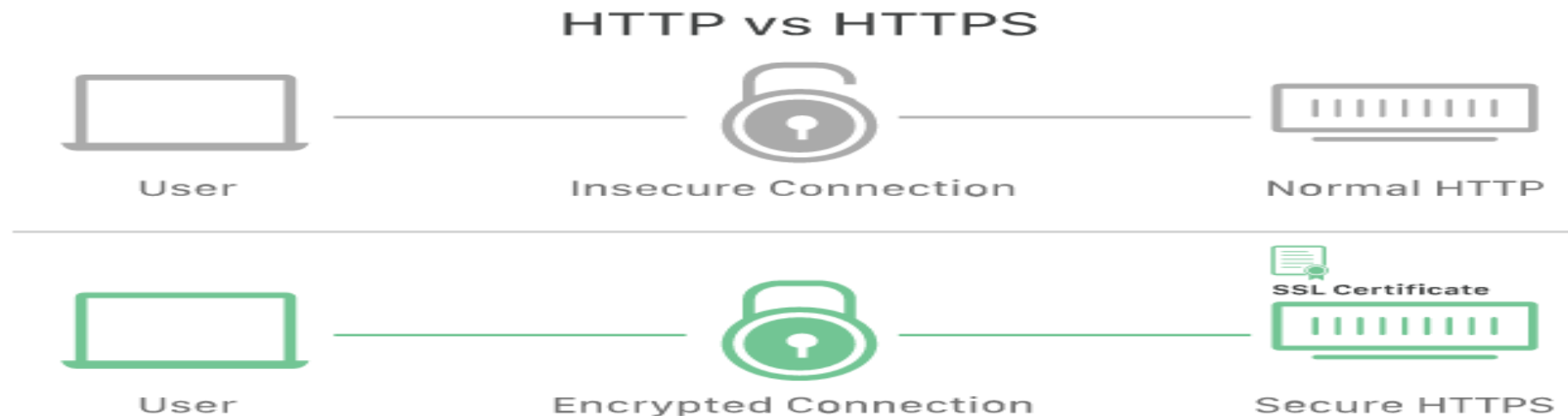
5.1 The internet and the world wide web

HTTP and HTTPS

HTTP: Stands for Hyper Text Transfer Protocol

HTTP is the set of rules for transferring files -- such as text, images, sound, video and other multimedia files -- over the web. As soon as a user opens their web browser, they are indirectly using HTTP.

HTTPS is HTTP with **encryption** and **verification**. The only difference between the two protocols is that HTTPS uses **TLS (SSL)** to encrypt normal HTTP requests and responses, and to digitally sign those requests and responses. As a result, HTTPS is far more secure than HTTP. A website that uses HTTP has http:// in its URL, while a website that uses HTTPS has https://.



5.1 The internet and the world wide web

Web Browsers

A web browser takes you anywhere on the internet. It retrieves information from other parts of the web and displays it on your desktop or mobile device. The information is transferred using the Hypertext Transfer Protocol, which defines how text, images and video are transmitted on the web. A web browser, or simply 'browser,' is an application used to access and view websites. Common web browsers include **Microsoft Edge, Internet Explorer, Google Chrome, Mozilla Firefox, and Apple Safari**

<https://www.youtube.com/watch?v=FxirRVJWUTs>



**BROWSER
BASICS**

5.1 The internet and the world wide web

Retrieval and location of web pages

HTML (HyperText Markup Language) is a language used to display content on browsers. All websites are written in HTML and hosted on a web server that has its own IP (**Internet Protocol**) address. When you type a web address in your browser, the browser looks at the DNS (**Domain Name Server**) to find the website's IP address before it can retrieve the website. is a system for finding IP addresses for a domain name given in a URL, The DNS process involves converting a URL (such as <https://www.epsomcollege.edu.my/>) into an IP address the computer can understand (such for example as 108.162.150.16).

How do you find your DNS server?

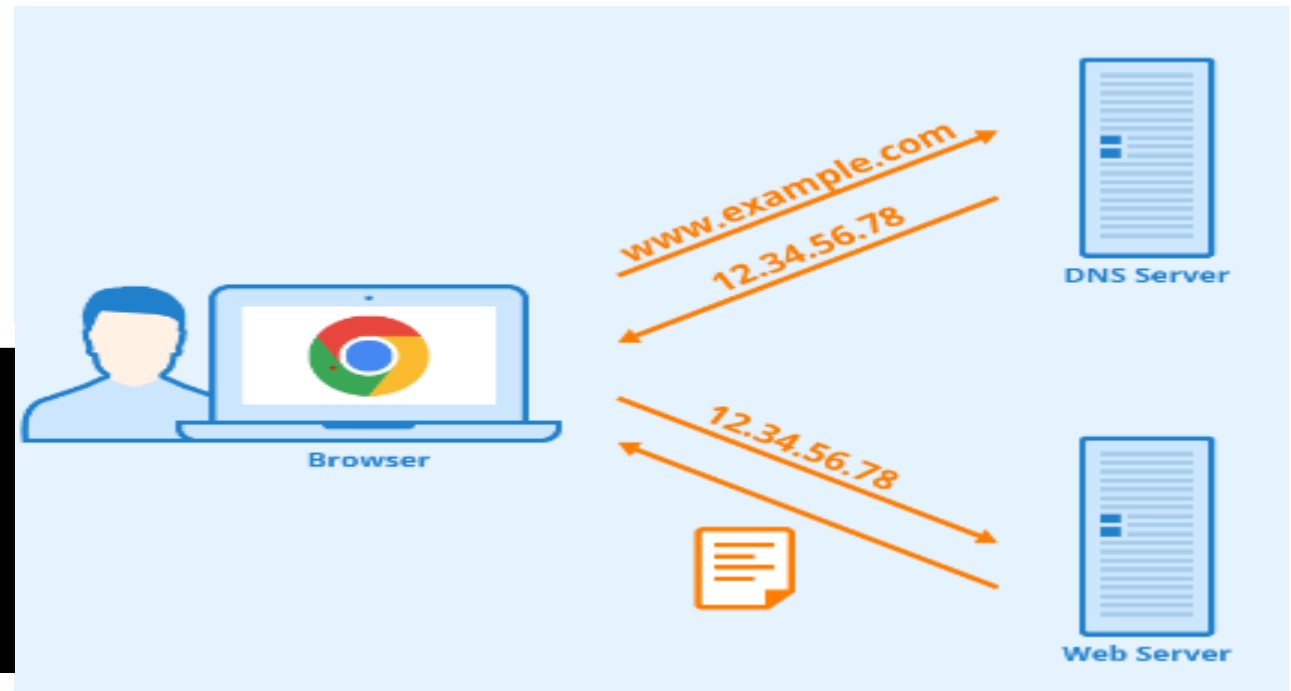
In Windows 10, right-click on the Start menu and click on Command Prompt (cmd).

Type “ipconfig /all”

cmd Command Prompt

```
Connection-specific DNS Suffix . : Broadcom 802.11n Network Adapter #10
Description . . . . . : 
Physical Address. . . . . : A8-B8-CF-19-66-92
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . : Yes
IPv6 Address. . . . . : 2001:e68:617c:4c01:809e:735c:ba7b:c4a5(Preferred)
Temporary IPv6 Address. . . . . : 2001:e68:617c:4c01:9d7c:72c0:c64e:ecbd(Preferred)
Link-local IPv6 Address . . . . : fe80::809e:735c:ba7b:c4a5%19(Preferred)
IPv4 Address. . . . . : 192.168.1.139(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, September 10, 2022 11:07:54 AM
Lease Expires . . . . . : Sunday, September 11, 2022 11:07:51 AM
Default Gateway . . . . . : fe80::aeb8:c6ff:fea0:30a9%19
                          192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 782810063
DHCPv6 Client DUID. . . . . : 00-01-00-01-1E-2A-78-BC-68-5B-35-80-17-A5
DNS Servers . . . . . : 2001:e68::b:68
                          2001:e68::b:69
                          192.168.1.1
                          0.0.0.0
NetBIOS over Tcpip. . . . . : Disabled
```

C:\Users\v90001010>



5.1 The internet and the world wide web

IP Address

An IP address, or Internet Protocol address, is a series of numbers that identifies any device on a network. such as sending mail, streaming video, or connecting to a website. Computers use IP addresses to communicate with each other both over the internet as well as on other networks.

The internet protocols manage the process of assigning each unique device its own IP address. (Internet protocols do other things as well, such as routing internet traffic.) This way, it's easy to see which devices on the internet are sending, requesting, and receiving what information.

Each device on the internet is given a unique address known as the INTERNET PROTOCOL (IP) ADDRESS. This is a 32-bit number which is usually written in the form: 192.168.1.1

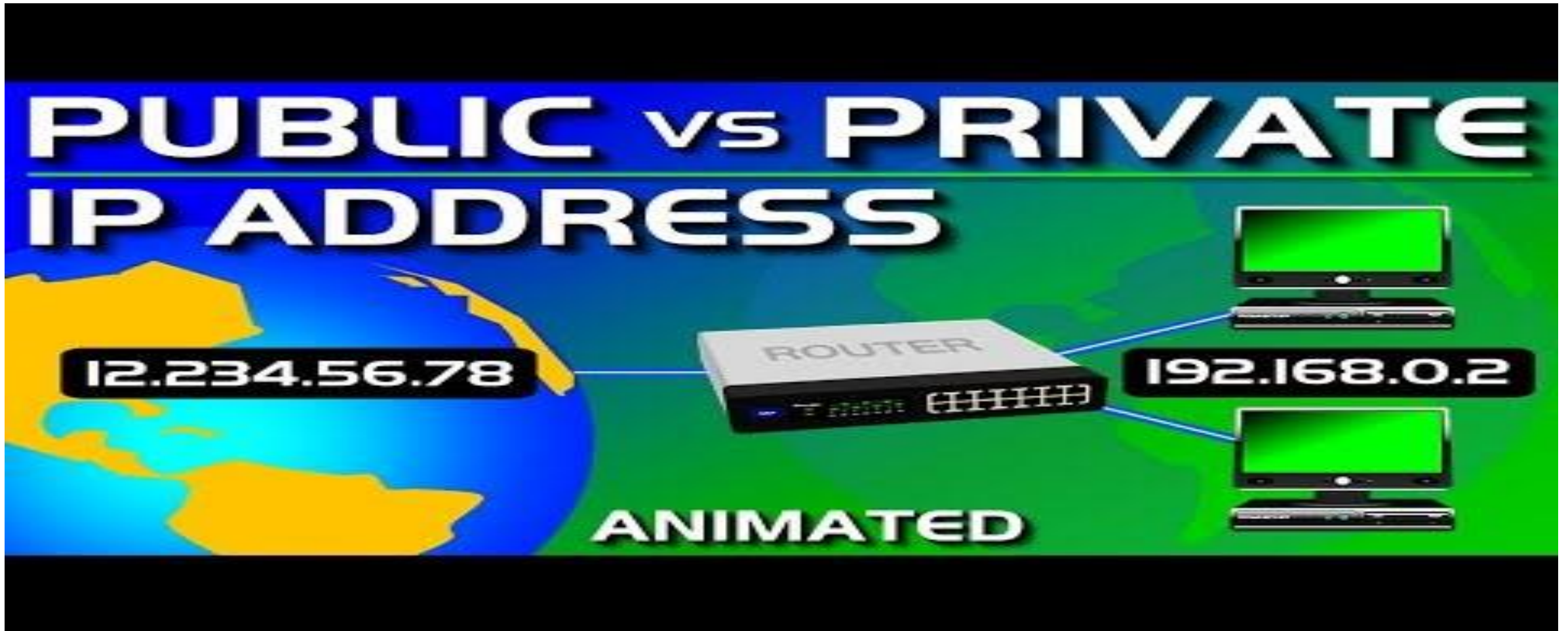
For window user: click command prompt – type: ipconfig/all

Activity: Find your IP address on your laptop or MacBook ?

5.1 The internet and the world wide web

IP Address

<https://www.youtube.com/watch?v=po8ZFG0Xc4Q>



5.1 The internet and the world wide web

Web Server:

A web server is **software and hardware that uses HTTP (Hypertext Transfer Protocol) and other protocols to respond to client requests made over the World Wide Web.** The main job of a web server is to display website content through storing, processing and delivering webpages to users. Nginx and Apache are undoubtedly the two most used web servers worldwide. Each of them holds about a third of the market

The main purpose of a web browser is to render hypertext markup language (HTML) and display web pages.

Functions include:

- storing bookmarks and favourites
- recording user history
- allowing use of multiple tabs
- storing cookies
- providing navigation tools
- providing an address bar

5.1 The internet and the world wide web

Web Server:

<https://www.youtube.com/watch?v=Yt1nesKi5Ec>



5.1 The internet and the world wide web

What is a Server?:

<https://www.youtube.com/watch?v=UjCDWCeHCzY>

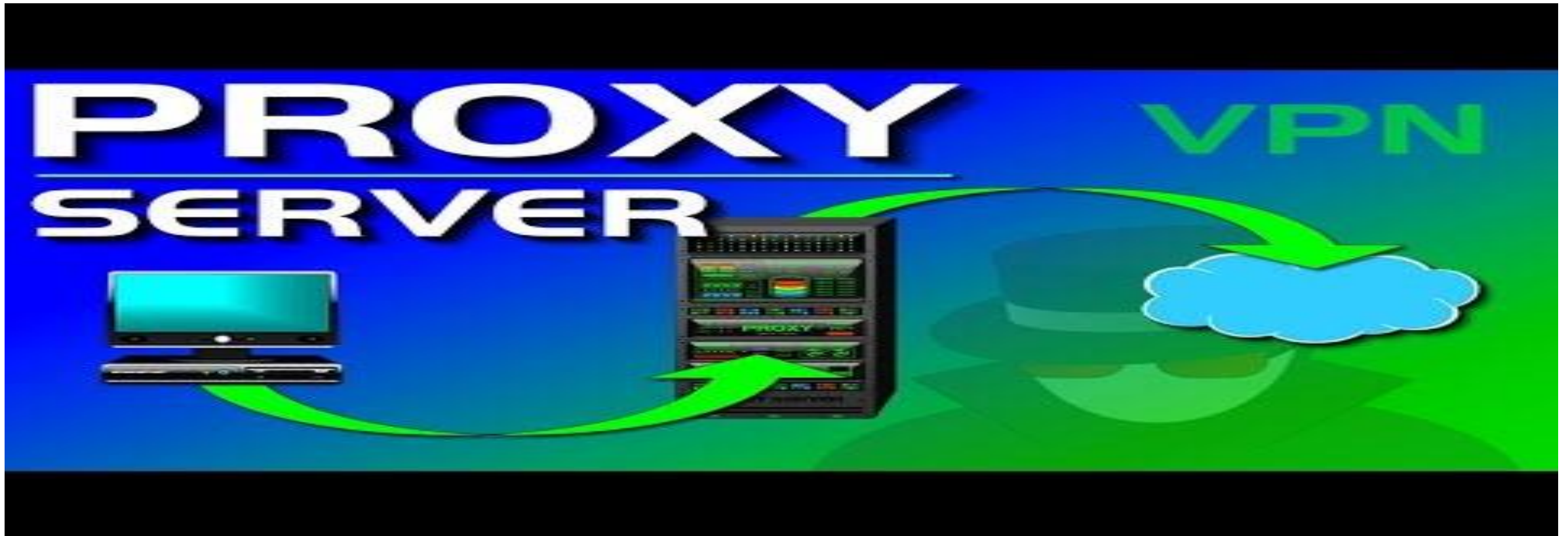


5.1 The internet and the world wide web

What is a Proxy Server?:

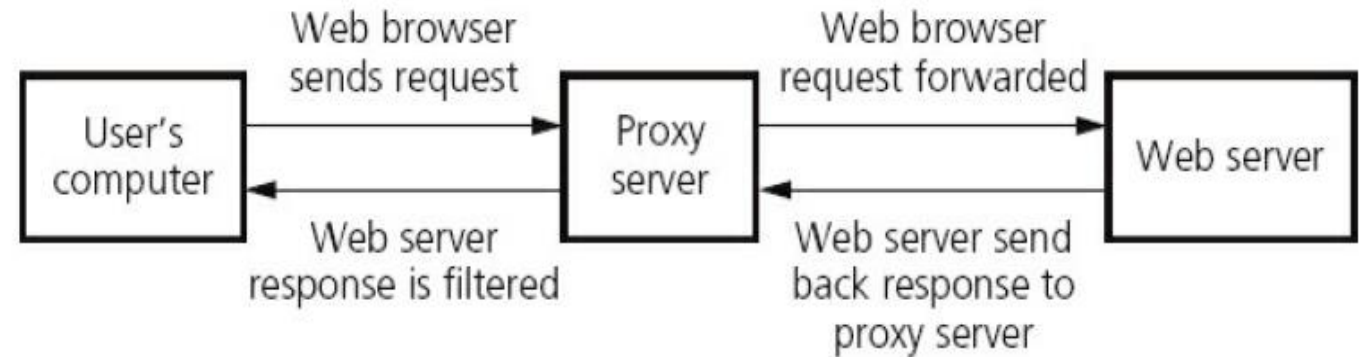
A proxy server is a system or router that **provides a gateway between users and the internet**. Therefore, it helps prevent cyber attackers from entering a private network. It is a server, referred to as an “intermediary” because it goes between end-users and the web pages they visit online.

<https://www.youtube.com/watch?v=5cPlukqXe5w>



5.1 The internet and the world wide web

PROXY SERVERS: act as an intermediary between the user and a web server:

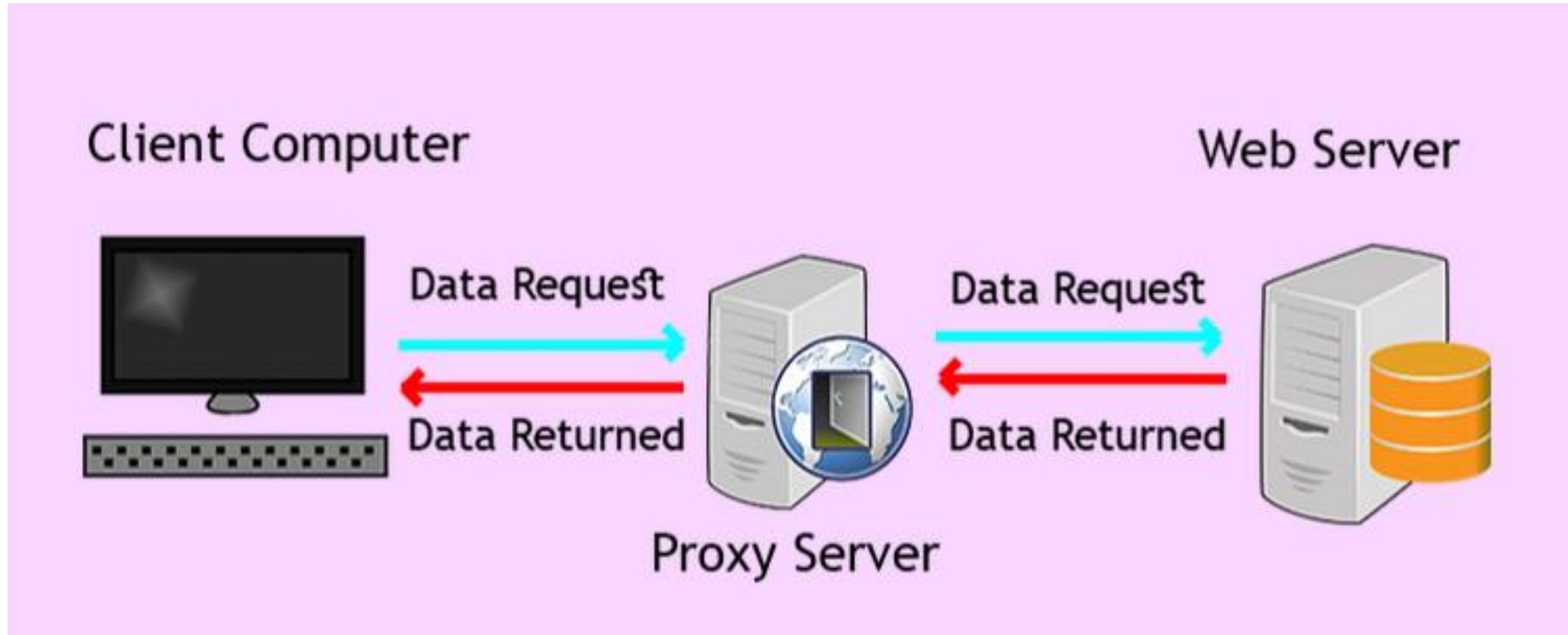


Functions of proxy servers include:

- allowing the internet 'traffic' to be filtered; they can block access to a website if necessary (similar type or reaction as a firewall)
- by using the feature known as a **CACHE**, they can speed up access to information from a website; when the website is first visited, the home page is stored on the proxy server; when the user next visits the website, it now goes through the proxy server cache instead, giving much faster access
- keeping the user's IP address secret – this clearly improves security
- acting as a firewall.

5.1 The internet and the world wide web

PROXY SERVERS: act as an intermediary between the user and a web server:



5.1 The internet and the world wide web

HTML (HyperText Markup Language):

HTML is the language for describing the structure of Web pages. HTML gives authors the means to: Publish online documents with headings, text, tables, lists, photos, etc. Retrieve online information via hypertext links, at the click of a button. HTML code ensures the proper formatting of text and images for your Internet browser. Without HTML, a browser would not know how to display text as elements or load images or other elements.

<https://www.youtube.com/watch?v=u0OeZfIfBRI>



HyperText Mark-up Language (HTML)

HYPertext MARK-UP LANGUAGE (HTML): is used when writing and developing web pages. HTML isn't a programming language but is simply a mark-up language. A mark-up language is used in the processing, definition and presentation of text (for example, specifying the colour of the text).

HTML uses **<tags>** which are used to bracket a piece of code; for example, **<html>** starts a standard cell in an HTML table, and **</html>** ends it. Whatever is between the two tags has been defined. Here is a short section of HTML code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h1>My First Heading</h1>
```

```
<p>My first paragraph.</p>
```

```
</body>
```

```
</html>
```

HyperText Mark-up Language (HTML)

HTML Structure and Presentation: When writing HTML code, it is very important to realise that there is a difference between the structure and the presentation.

- **STRUCTURE:** is the essential part of the HTML document; it includes the semantics (the meaning of a word, phrase, or text) and structural mark-up of the document.
- **PRESENTATION:** is the style of the document; i.e. how the document will look (or even sound if it includes multimedia elements).
- HTML document (which contains the structure and the actual content) and a separate **CSS (CASCADING STYLE SHEET)** file. The css file will contain everything to control the actual presentation of the web page.

HyperText Mark-up Language (HTML)

What is CSS?

- Cascading Style Sheets (CSS) is used to format the layout of a webpage.
- With CSS, you can control the color, font, the size of text, the spacing between elements, how elements are positioned and laid out, what background images or background colors are to be used, different displays for different devices and screen sizes, and much more!

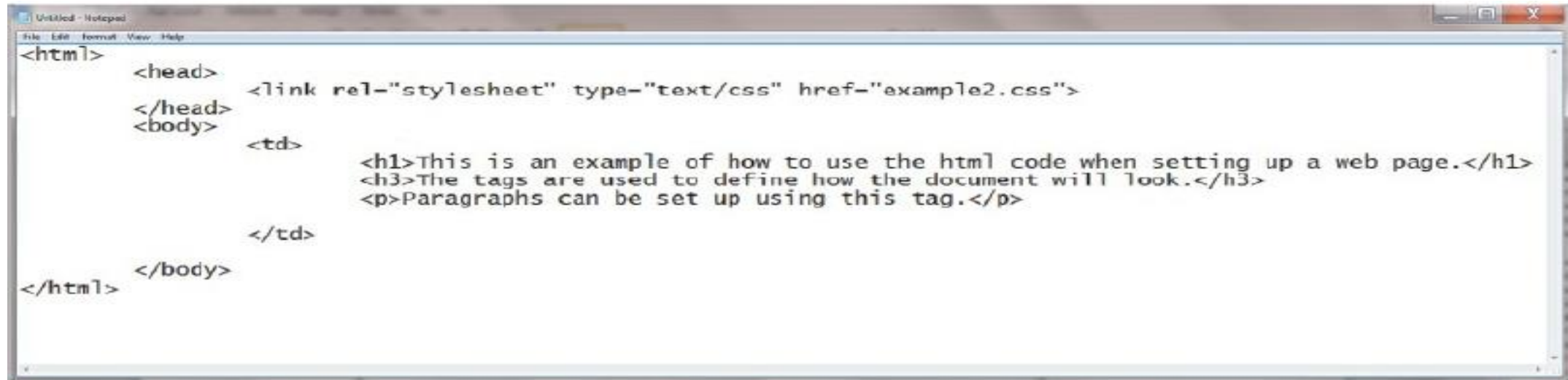
Example 1:

- `<!DOCTYPE html>`
- `<html>`
- `<body>`
- `<h1 style="color:blue;">A Blue Heading</h1>`
- `<p style="color:red;">A red paragraph.</p>`
- `</body>`
- `</html>`

HyperText Mark-up Language (HTML)

- CSS - Example 2:

This shows how the css file (example2.css) is created for use in the document below; h1, h3 and p have all been defined, so when the file is attached below, a web browser knows how to display this web page.

A screenshot of a Notepad window titled 'Untitled - Notepad'. The window contains the following HTML code:

```
<html>
  <head>
    <link rel="stylesheet" type="text/css" href="example2.css">
  </head>
  <body>
    <td>
      <h1>This is an example of how to use the html code when setting up a web page.</h1>
      <h3>The tags are used to define how the document will look.</h3>
      <p>Paragraphs can be set up using this tag.</p>
    </td>
  </body>
</html>
```

5.1 The internet and the world wide web

Cookies:

<https://www.youtube.com/watch?v=rdVPfIECed8>



5.1 The internet and the world wide web

Retrieval and location of web pages

Cookies:

Are small pieces of text sent to your browser by a website you visit. They help that website remember information about your visit, which can both make it easier to visit the site again and make the site more useful to you. Cookies are used for functions, including: Saving personal details, tracking user preferences, holding items in an online shopping cart, storing login details

There are two types of cookie:

- » Session cookie
- » Persistent (or permanent) cookie.

Session cookies: are used, for example, when making online purchases. Session cookies do not retain any information on your device or send information from your device. These cookies are deleted when the session expires or is terminated when the browser window is closed.

Persistent cookies: remain on the device until you erase them or they expire. Many countries have introduced laws to protect users and these cookies are deactivated after six months.

5.1 The internet and the world wide web

Activity 5.1

- 1 A URL being entered is: **http://www.urlexample.co.ie/sample_page**
Identify:
 - a the domain name
 - b the domain type
 - c the file name
 - d which protocol is being used.
- 2
 - a Give two differences between session cookies and persistent cookies.
 - b Describe three uses of cookies.
- 3 The following table shows five features of the internet and the World Wide Web. Tick (✓) the appropriate box to indicate which feature refers to the internet and which feature refers to the World Wide Web:

Feature	Internet	World Wide Web
it is possible to send and receive emails		
makes use of http protocols		
uses URLs to specify the locations of websites and web pages		
resources can be accessed by using web browsers		
makes use of TCP and IP		

- 4 Why do you think persistent cookies are sometimes referred to as **tracking cookies**? Give at least two pieces of evidence to support your answer.

5.1 The internet and the world wide web

- 2 A user wishes to locate a website so they can download some documents. The following sequence shows six steps in the location of the website and the download of the required documents. However, the sequence of steps shown is not in the correct order.

By writing the numbers 1 to 6, put each step in its correct order.

Order	Description of step
	The DNS server 1 cannot find the required website in its database or cache and sends out a request to DNS server 2
	The IP address is then sent back to the user's computer
	The computer now sets up a communication with the website server and the required pages are downloaded
	User opens their web browser and types in the URL; the web browser asks DNS server 1 for the IP address
	User's browser interprets HTML and displays the web pages on the user's computer
	DNS server 2 finds the URL and sends the IP address back to DNS server 1 which puts the IP address and URL into its database and cache

5.1 The internet and the world wide web

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5.1 The internet and the world wide web

- 1 a Explain the fundamental differences between the internet and the world wide web (www).

.....

.....

.....

.....

A user typed in: **https://www.hoddereducation.com/comp_science_sample**

Identify:

- i the protocol being used:
- ii the domain host:
- iii the domain type:
- iv the file name:

5.1 The internet and the world wide web

c Describe **three** of the features of web browsers.

1

.....

.....

2

.....

.....

3

.....

.....

5.1 The internet and the world wide web

IGCSE Past Exam -Question 1

(d) Customers will use a web browser to access Victoria's website.

Victoria writes a paragraph of text to explain how the website will be displayed on a customer's computer.

Use the list given to complete Victoria's paragraph by inserting the correct **six** missing terms. Not all terms will be used.

- browser
- domain name
- firewall
- hexadecimal
- HTML
- https
- MAC address
- search engine
- Uniform Resource Locator (URL)
- web server

5.1 The internet and the world wide web

IGCSE Past Exam -Question 1

The user enters the website into the address bar.

The protocol that is used is The URL contains

the for the website. This is used to look up the

IP address of the company. A DNS server stores an index of IP addresses.

The browser sends a request to the as this is

where the files for the website are stored. The files are sent back to the

..... as files.

5.1 The internet and the world wide web

IGCSE Past Exam -Question 2

(e) When customers access Victoria's website they will be given the message:

This website uses cookies. An explanation of their purpose can be found in our cookies policy.

(i) Explain what is meant by cookies.

.....

.....

.....

.....

(ii) Explain why Victoria would use cookies as part of her website.

.....

.....

.....

.....

.....

5.1 The internet and the world wide web

IGCSE Past Exam -Question 3

Priya creates a website to sell her old comic books and superhero figures.

- (a) She uses HTML to create her website. The HTML she produces has both structure and presentation.

Explain what is meant by HTML **structure** and **presentation**. Include an **example** of each.

Structure

.....
.....
.....

Presentation

.....
.....
.....

[4]

5.1 The internet and the world wide web

IGCSE Past Exam -Question 4

(b) Priya uses cookies in her website.

Five statements are given about cookies.

Tick (✓) to show if the statement is **True** or **False**.

Statement	True (✓)	False (✓)
Cookies can be used to store a customer's credit card details		
Cookies can be used to track the items a customer has viewed on a website		
Cookies will corrupt the data on a customer's computer		
Cookies are downloaded onto a customer's computer		
Cookies can be deleted from a customer's computer		

[5]

5.1 The internet and the world wide web

IGCSE Past Exam -Question 5

Thomas has an online business that sells homemade furniture. He has a web server that hosts his website for his business.

- (a) Describe the role of a web browser in requesting and displaying the web pages for the website.

.....

..... [3]

5.1 The internet and the world wide web

IGCSE Past Exam -Question 6

The table shows **four** definitions.

Complete the table giving the missing **Term** for each definition.

Term	Definition
	A data transmission method that sends data one bit at a time, down a single wire
	An address given to a device on a network. The address is assigned by the network
	The software used to render HTML and display a web page
	An address given to a device at the manufacturing stage that can be used to identify the device on a network

- IP address
- MAC address
- Serial (simplex)
- Browser

5.1 The internet and the world wide web

IGCSE Past Exam -Question 7

Meena uses a browser to research information for her business.

(a) Give **three** functions of a browser.

Any **three** from:

1

2

3

[3]

5.1 The internet and the world wide web

IGCSE Past Exam -Question 8

(b) The URL for the music company's website is:

`https://www.rockict.net/index.htm`

Part 1 Part 2

(i) Identify what **Part 1** and **Part 2** represent in this URL.

Part 1

Part 2 [2]

(ii) Describe what is meant by **https**.

.....

..... [2]

5.1 The internet and the world wide web

IGCSE Past Exam -Question 8

(c) When a customer enters the website, a message is displayed:

"RockICT makes use of cookies. By continuing to browse you are agreeing to our use of cookies."

Explain why the music company uses cookies.

.....	
.....	
.....	
.....	

[2]

(d) The music company is concerned about the security of its website.

The company uses a proxy server as part of its security system.

Describe the role of a proxy server in the security system.

.....	
.....	
.....	
.....	

5.1 The internet and the world wide web

IGCSE Past Exam -Question 9

Describe the role of an Internet Service Provider (ISP).

.....

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5.1 The internet and the world wide web

IGCSE Past Exam -Question 10

Kamil correctly answers an examination question about a number of internet terms.

Six different terms have been removed from Kamil's answer.

Complete the sentences in Kamil's answer, using the list given. Not all terms in the list need to be used.

- browser
- connection
- domain name server (DNS)
- Internet
- Internet Service Provider (ISP)
- IP address
- MAC address
- network
- protocol
- uniform resource locator (URL)
- webpages
- hypertext mark-up language (HTML)

|

5.1 The internet and the world wide web

IGCSE Past Exam -Question 10

A is a program that allows a user to view

An is a company that provides a connection to access the

The main that governs the transmission of data using the Internet is http.

The is provided by the network, and given to each device on the network.

[6]

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 1

The user enters the websiteURL..... into the address bar.

The protocol that is used ishttps..... . The URL contains theDomain name..... for the website. This is used to look up the IP address of the company. A DNS server stores an index of IP addresses.

The browser sends a request to theWeb server..... as this is where the files for the website are stored. The files are sent back to theBrowser..... asHTML..... files.

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 2

(e) When customers access Victoria's website they will be given the message:

This website uses cookies. An explanation of their purpose can be found in our cookies policy.

(i) Explain what is meant by cookies.

- – Small packets of data
- – that are stored by the web browser

(ii) Explain why Victoria would use cookies as part of her website.

Four from:

- – To store a customer's password
- – To store a customer's credit card details
- – so they do not need to be re-entered in future
- – To track what the customer has viewed on the website
- – So she can send them adverts that match their preferences

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 3

Structure

- This is the layout of the web page
- e.g. placing an image alongside some text // example of tag, such as <div>

Presentation

- This is the formatting/style of the web page
- e.g. the colour that is applied to some text // example of tag, such as <font-color>

4

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 4

1 mark per each correct row.

5

Statement	True (✓)	False (✓)
Cookies can be used to store a customer's credit card details	✓	
Cookies can be used to track the items a customer has viewed on a website	✓	
Cookies will corrupt the data on a customer's computer		✓
Cookies are downloaded onto a customer's computer	✓	
Cookies can be deleted from a customer's computer	✓	

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 5

Thomas has an online business that sells homemade furniture. He has a web server that hosts his website for his business.

- (a) Describe the role of a web browser in requesting and displaying the web pages for the website.

..... Any three from:
..... – Sends request to webserver
..... – Receives web pages back from webserver
..... – Converts HTML to display web page
..... – Manages protocols

.....
..... [3]

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 6

The table shows **four** definitions.

Complete the table giving the missing **Term** for each definition.

Term	Definition
Serial (ignore any ref to simplex etc.)	A data transmission method that sends data one bit at a time, down a single wire
IP (address)	An address given to a device on a network. The address is assigned by the network
Browser	The software used to render HTML and display a web page
MAC (address)	An address given to a device at the manufacturing stage that can be used to identify the device on a network

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 7

Meena uses a browser to research information for her business.

(a) Give **three** functions of a browser.

Any three from:

- | | | |
|---------|---------------------------------------|-------|
| 1 | - Convert HTML code | |
| | - Display web pages | |
| 2 | - Check if a website is secure | |
| | - Request web pages from a web server | |
| 3 | - Send URL/domain name | |
| | - Runs active script | |
| | - Store history/favourites/bookmarks | |
| | - Create tabs | |

[3]

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 8

(b) The URL for the music company's website is:

https://www.rockict.net/index.htm

Part 1 Part 2

(i) Identify what **Part 1** and **Part 2** represent in this URL.

Part 1 domain (name)

Part 2 file name/webpage name [2]

(ii) Describe what is meant by **https**.

Any two from:

- Hypertext Transfer Protocol Secure // it is the access protocol // It is a protocol
- It means the website uses SSL/TLS
- It means data sent (to and from the webserver) is encrypted

.....
..... [2]

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 8

(c) When a customer enters the website, a message is displayed:

“RockICT makes use of cookies. By continuing to browse you are agreeing to our use of cookies.”

Explain why the music company uses cookies.

Any two from e.g. :

- | | | |
|-------|---|-------|
| | - To store items that a customer has added to an online shopping basket | |
| | - To store a customer's credit card details | |
| | - To store log-in details | |
| | - To track what product a customer browses // Track music preferences | |
| | - Targeted advertising // making recommendations | |
| | - Personalises/customises the experience | |
| | - Shows who are new and returning customers | |
| | - To speed up log-in times | |
| | - To speed up/allow single click purchases | |
| | - Improves the experience | [2] |

(d) The music company is concerned about the security of its website.

The company uses a proxy server as part of its security system.

Describe the role of a proxy server in the security system.

Any four from:

- Prevents direct access to the webserver // Sits between user and webserver
- If an attack is launched it hits the proxy server instead // can be used to help prevent DDOS // help prevent hacking of webserver
- Used to direct invalid traffic away from the webserver
- Traffic is examined by the proxy server // Filters traffic
- If traffic is valid the data from the webserver will be obtained by the user
- If traffic is invalid the request to obtain data is declined
- Can block requests from certain IP addresses

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 9

13 Describe the role of an Internet Service Provider (ISP).

.... Any six from e.g.:

- Provide access to the internet / dial up / broadband
- Usually charge a monthly fee
- Monitor usage
- Give users an IP address
- Determine bandwidth
- Supports domain names
- Provide security services
- Provide web hosting facilities
- Provide access to Email / Mailbox
- Provides online data storage

5.1 The internet and the world wide web

IGCSE Past Exam -Answer 10

A **browser** is a program that allows a user
to view **webpages**

An **Internet Service Provider (ISP)** is a company that provides a
connection to access the **Internet**

The main **protocol** that governs the
transmission of data using the Internet is http.

The **IP address** is provided by the network,
and given to each device on the network.

[6]

Learning Objectives

5.2 Digital currency

Candidates should be able to:

- 1 Understand the concept of a digital currency and how digital currencies are used
- 2 Understand the process of blockchain and how it is used to track digital currency transactions

Notes and guidance

- A digital currency is one that only exists electronically
- Blockchain, in its basic form, is a digital ledger, that is a time-stamped series of records that cannot be altered

5.2 Digital currency

What is digital currency?

<https://www.youtube.com/watch?v=yGjbe1ryZAI>



5.2 Digital currency

What is digital currency?

As the digital economy continues to expand, the use of digital currency is also increasing rapidly. Digital currency is money in an electronic form exchanged for goods and services without the use of physical money such as paper bills or coins.

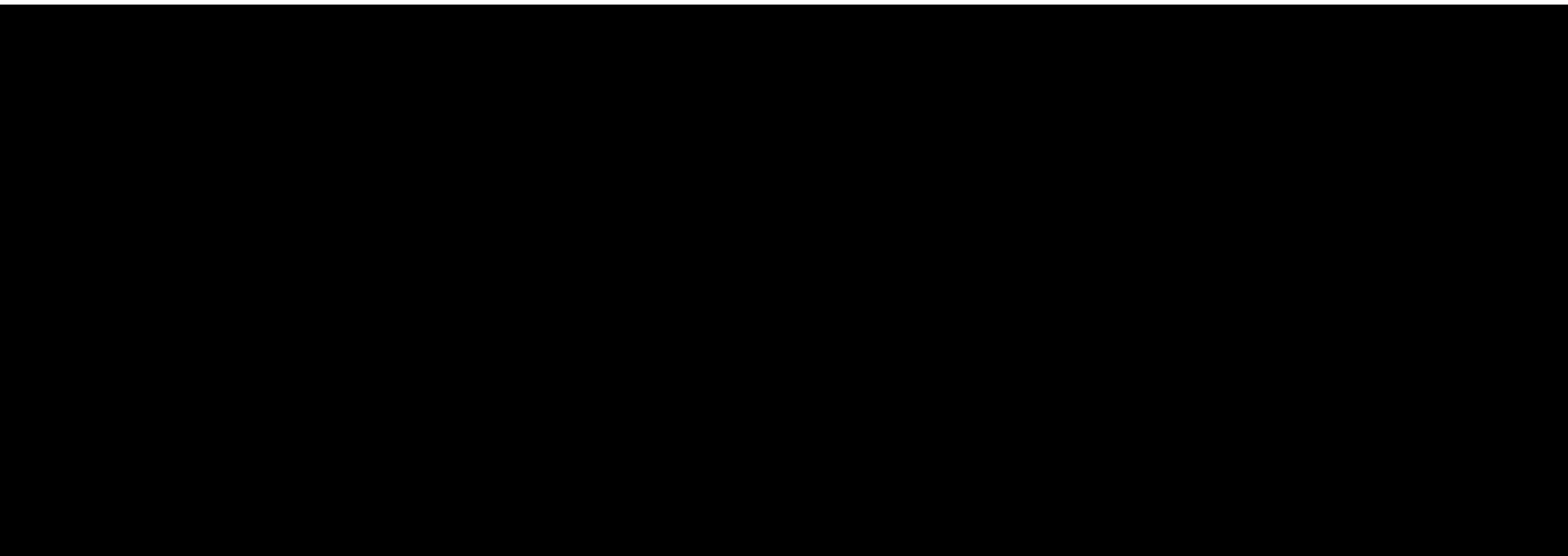
Digital money today mostly facilitates the movement of **fiat currency** (mean money issued and backed by a government such as the U.S. dollar, Malay Ringi, or the euro) and Digital currency relies on a **central banking system**. In a centralized process, a system of serial numbers is usually used to make sure each note is unique. Bank partners are used to distribute cash into the economy But digital currency now also includes cryptocurrencies. Bitcoin is the original cryptocurrency and was privately developed as a means of exchange on the internet and **cryptocurrency** is not centralized system. **Cryptography** is often used to make user identities and transaction details anonymous. A digital currency can also bypass bank and financial institution intermediaries and be provided directly to users. The cryptocurrency system works by being within a **blockchain** network which means it is much more secure.

5.2 Digital currency

Blockchain:

What is Blockchain??

<https://www.youtube.com/watch?v=8o9QxMxhTp8>



5.2 Digital currency

Blockchain:

<https://www.youtube.com/watch?v=sTFZras-1Lo>



5.2 Digital currency

Blockchain:

How does a blockchain work?

https://www.youtube.com/watch?v=SSo_ElwHSd4



Blockchain

— *Simply explained* —

5.2 Digital currency

Blockchain:

<https://www.youtube.com/watch?v=kHybf1aC-jE>

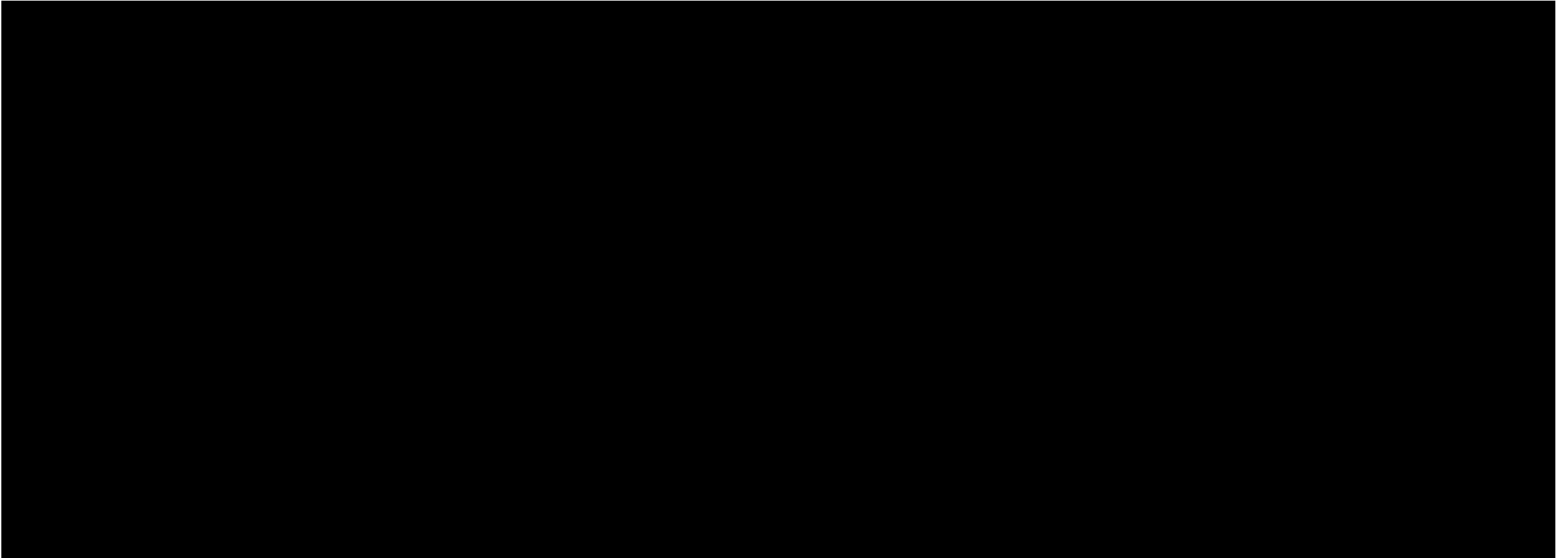


5.2 Digital currency

Blockchain:

What is Encryption and How Does it Work?

<https://www.youtube.com/watch?v=r4HQ8Bp-pfw>



4. 5.2 Digital currency:

Practical Encryption 1: Open Python IDLE and Enter the code below:

 encryptanddecrypt.py - C:/Users/v90001010/encryptanddecrypt.py (3.10.6)

File Edit Format Run Options Window Help

```
alphabets="abcdefghijklmnopqrstuvwxyz"
message=input("Enter a message to encrypt: ")
encrypt,decrypt= "", ""
key = 10

for letter in message:
    new_position=(alphabets.find(letter)+key)%len(alphabets)
    encrypt+=alphabets[new_position]

for letter in encrypt:
    new_position=(alphabets.find(letter)-key)%len(alphabets)
    decrypt+=alphabets[new_position]

print("Encrypt message:",encrypt)
print("Decrypt message:",decrypt)
```

```
===== RESTART: C:/Users/v90001010/encrypt
Enter a message to encrypt: noureddine
Encrypt message: xyebonnsox
Decrypt message: noureddine
```


5.2 Digital currency

Blockchain:

What is Hashing?

<https://www.youtube.com/watch?v=cczlpiiu42M>



Passwords & hash functions

— *Simply explained* —

5.2 Digital currency

Blockchain:

What is Hashing?

<https://www.youtube.com/watch?v=2BldESGZKB8>



What is Hashing?



5.2 Digital currency

Blockchain:

Encryption vs Hashing

<https://www.youtube.com/watch?v=Sre8D9X1BzE>



5.2 Digital currency

Blockchain:

Blockchain defined:

Blockchain is a decentralised database. Essentially, the blockchain consists of a number of interconnected computers but they are not connected to a central server. All transaction data is stored on all computers in the blockchain network.

Blockchain is a shared, immutable (**unchanging over time or unable to be changed**) ledger that facilitates the process of recording transactions and tracking assets in a business network. An **asset** (**for example: house, car, cash, land**) can be tangible (**what is a tangible is something that's literally tangible can be touched for example a house**) or intangible (**intellectual property, patents, copyrights, branding**). Virtually anything of value can be tracked and traded on a blockchain network, reducing risk and cutting costs for all involved.

Why blockchain is important:

Business runs on information. The faster it's received and the more accurate it is, the better. Blockchain is ideal for delivering that information because it provides immediate, shared and completely transparent information stored on an immutable ledger that can be accessed only by permissioned network members. A blockchain network can track orders, payments, accounts, production and much more. And because members share a single view of the truth, you can see all details of a transaction end to end, giving you greater confidence, as well as new efficiencies and opportunities.

What is a ledger? A ledger is a book or collection of accounts in which account transactions are recorded

5.2 Digital currency

Blockchain:

Blockchain defined:

.Why blockchain is important:

Whenever a new transaction takes place, all the networked computers get a copy of the transaction; therefore **it cannot be changed without the consent** of **all** the network members. This effectively removes the risk of security issues such as hacking. Blockchain is used in many areas, such as: _

- » cryptocurrency (digital currency) exchanges
- » smart contracts
- » research (particularly within pharmaceutical companies)
- » politics
- » education.

Distributed ledger technology (DLT): is a digital system for recording the transaction of assets in which the transactions and their details are recorded in multiple places at the same time.

5.2 Digital currency

Blockchain:

Blockchain defined:

.Why blockchain is important:

The timestamp or timestamp: is a small data stored in each block as a unique serial and whose main function is to determine the exact moment in which the block has been mined and validated by the network.

In the context of verifying the authenticity and creation of data, Nakamoto developed **timestamps** based on the use of hash-based **proof of work**. In this network, a **Timestamp Server** takes the **hash** of a block containing transactions to be timestamped and published on the network. By doing so, the server ensures that the transactions have existed at this point in time. This system is chronologically enforced as each transaction **hash** contains the timestamp of the previous block of transactions, forming a chain of blocks.

Mining is the process that Bitcoin and several other cryptocurrencies use to generate new coins and verify new transactions.

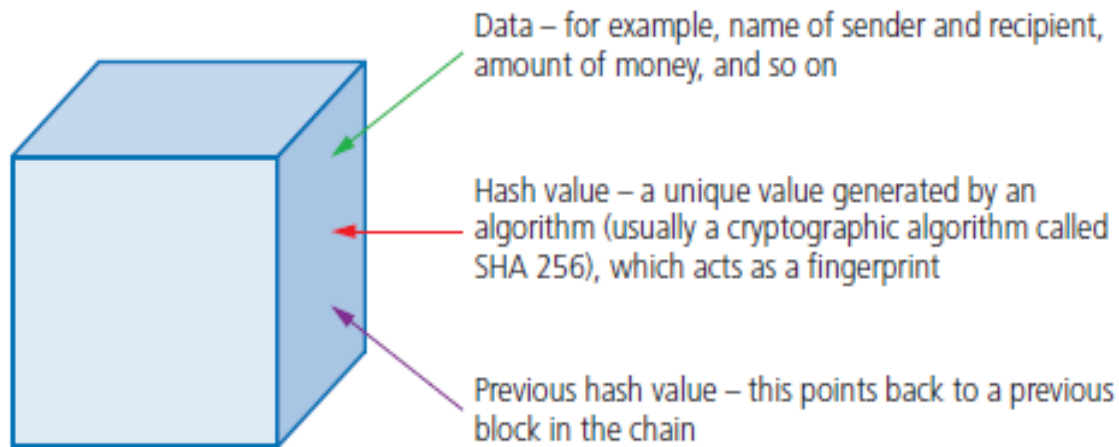
A **Genesis Block** is the name given to the first block a cryptocurrency, such as Bitcoin, ever mined. A blockchain consists of a series of so-called blocks that are used to store information related to transactions that occur on a blockchain network.

<https://academy.binance.com/en/glossary/genesis-block>

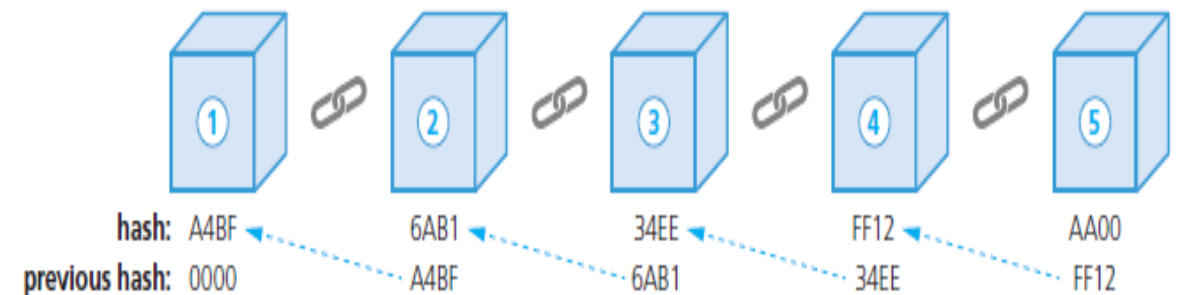
5.2 Digital currency

Blockchain:

Whenever a new transaction takes place, a new **block** is created:

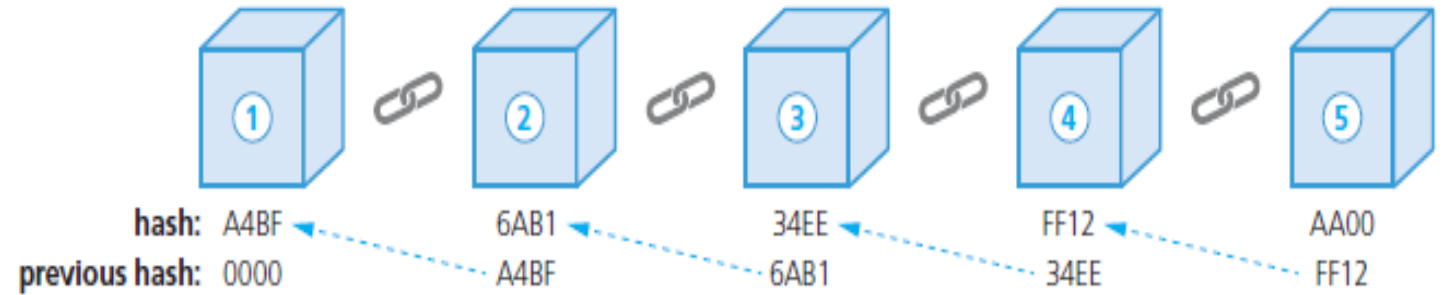


A new hash value is created each time a new block is created. This hash value is unique to each block and includes a **timestamp**, which identifies when an event actually takes place. We will now consider what happens when a chain of blocks is created. Figure 2 shows part of a typical blockchain:



5.2 Digital currency

Blockchain:

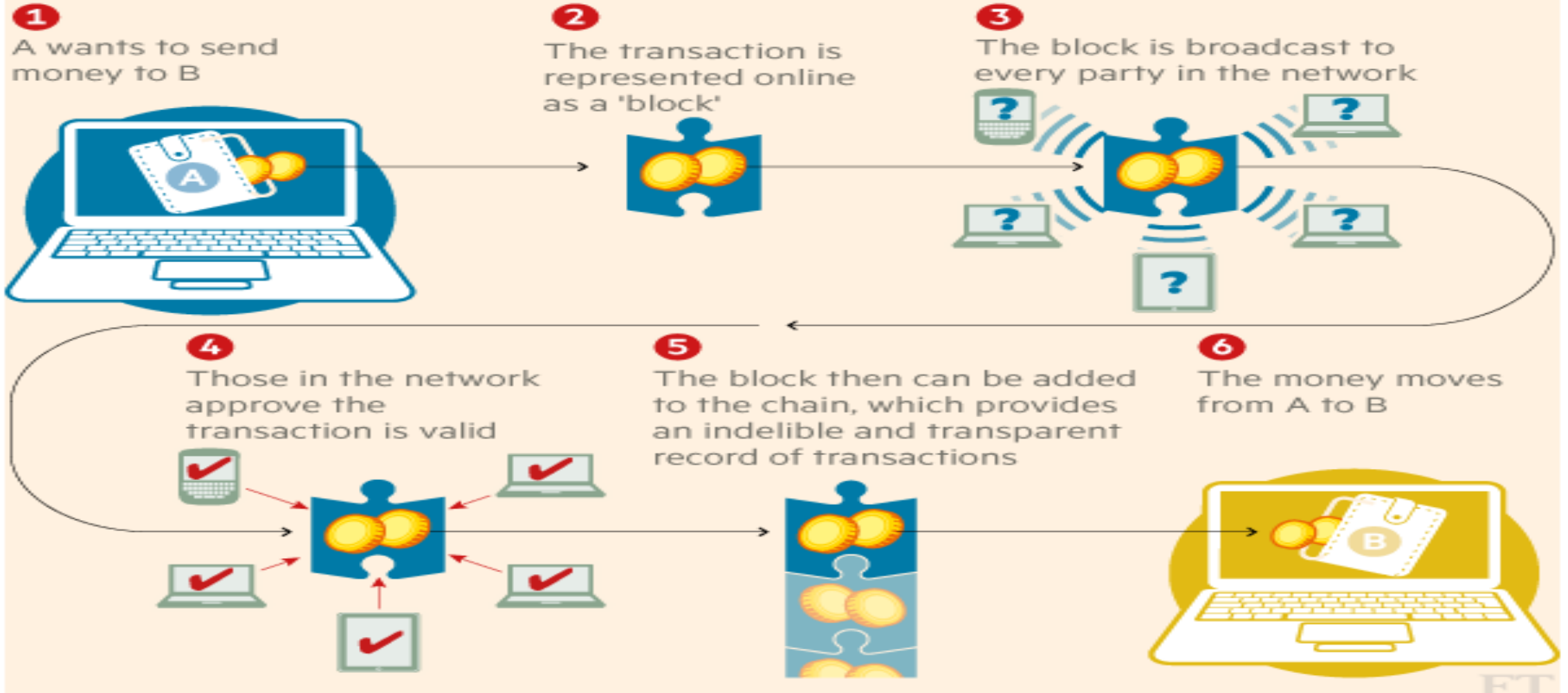


It is clear from Figure above how these blocks are connected. Block '1' is known as the **genesis block** since it doesn't point to any previous block. Now suppose block '2' is changed in some way. Any changes to the data within block '2' will cause the value of the hash to change (it will no longer have the value 6AB1). This means that block '3' and beyond will now be invalid since the chain was broken between block '2' and '3' (previous hash 6AB1 in block '3' is no longer valid). This will prevent tampering (for example, by a hacker). However, it may be crossing your mind that computers are now so fast that it should be possible to quickly create a whole new string of blocks, and therefore recreate a new chain before the problem has been discovered. This is prevented by **proof-of-work**, which makes sure it takes ten minutes to determine the necessary proof-of-work for *each* block **before** it can be added to the chain. This is 'policed' by **miners**, which are special network users that get a commission for each new block created. Thus, the whole process of creating new blocks is slowed down which foils hackers and also means that the currency is regulated by all the network computers. Consequently, this makes it almost impossible to hack into the blockchain since it would be necessary to attack every single block in the chain at the same time. It only takes one block to break the link for any transaction to be terminated. When a new block is created, it is sent to each computer in the blockchain and is checked for correctness before being added to the blockchain. If a new network user is created, they get a copy of everything in the whole blockchain system.

5.2 Digital currency

Blockchain:

How a blockchain works



5.2 Digital currency

Blockchain: Question and answer 1:

Q 1. What do you know about Blockchain? What is the difference between Bitcoin blockchain and Ethereum blockchain?

Answer 1:

The blockchain is a **decentralized distributed database** of immutable records. The technology was discovered with the invention of Bitcoins(the first cryptocurrency). It's a trusted approach and there are a lot of companies in the present scenario which are using it. As everything is secure, and because it's an open-source approach, it can easily be trusted in the long run.

Topics	Bitcoin	Ethereum
<i>Concept</i>	Digital Currency	Smart Contracts
<i>Founder</i>	Satoshi Nakamoto	Vitalik Buterin
<i>Release Method</i>	Genesis Block Mined	Presale
<i>Cryptocurrency Used</i>	Bitcoin(Satoshi)	Ether
<i>Algorithm</i>	SHA-256	Ethash
<i>Blocks Time</i>	10 Minutes	12-14 Seconds
<i>Scalable</i>	Not yet	Yes

Bitcoin Blockchain and Ethereum Blockchain

5.2 Digital currency

Blockchain: Question and answer 1:

Q 2. What is the principle on which blockchain technology is based on?

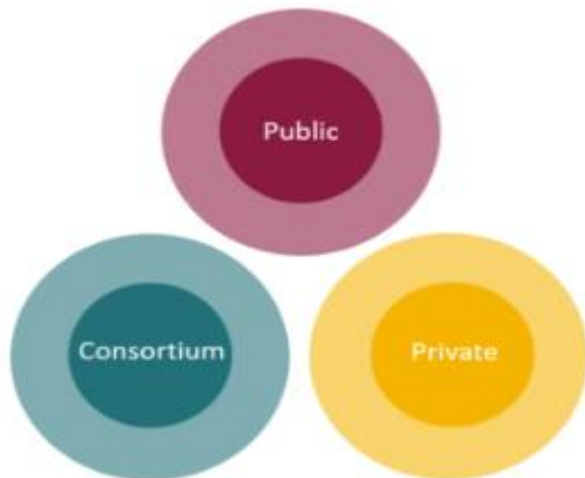
Answer 2:

It enables the information to be distributed among the users without being copied.

Q 3. What are the different types of Blockchains?

Answer 3:

Blockchains are of three types



5.2 Digital currency

Blockchain: Question and answer 1:

Q 4. Why is Blockchain a trusted approach?

Answer 4:

- Blockchain can be trusted due to so many reasons.
- Its compatibility with other business applications due to its open-source nature.
- Its security. As it was meant for online transactions, the developers have paid special attention to keeping up the pace when it comes to its security.
- It really doesn't matter what type of business one owns, Blockchain can easily be considered.

Q 5. Name the two types of records that are present in the blockchain database?

Answer 5:

These records are block records and transactional records. Both these records can easily be accessed, and the best thing is, it is possible to integrate them with each other without following the complex algorithms.

Q 6. What are the properties of Blockchain?

Answer 6:

There are four key features of blockchain:

- Decentralized Systems
- Distributed ledger
- Safer & Secure Ecosystem
- Minting (make a coin) by stamping metal)

5.2 Digital currency

Blockchain: Question and answer 1:

Q 7. What is encryption? What is its role in Blockchain?

Answer 7:

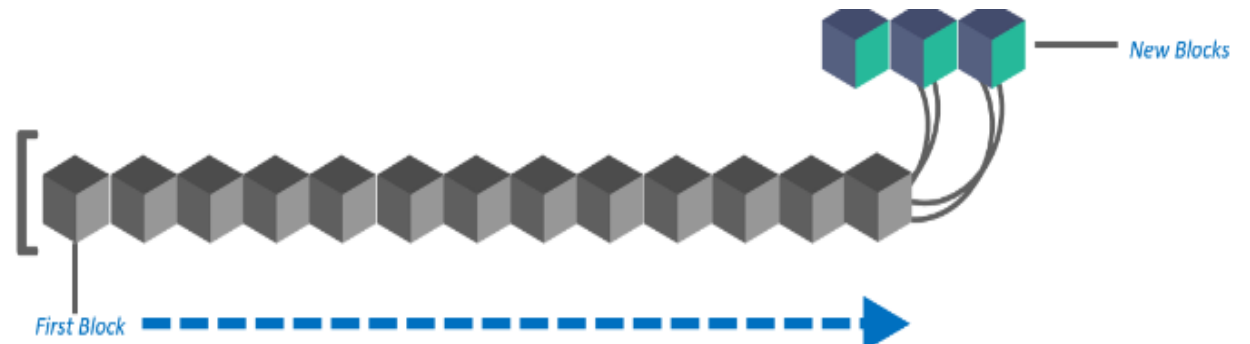
Data security always matters. Encryption is basically an approach that helps organizations to keep their data secure.



The encrypted data is encoded or changed up to some extent before it is sent out of a network by the sender and only authorized parties can access that information. In Blockchain, this approach is useful because it simply adds more to the overall security and authenticity of blocks and helps to keep them secure.

Q 8. What do you mean by blocks in blockchain technology?

Ans 8: Blockchain consists of a list of records. Such records are stored in blocks. These blocks are in turn linked with other blocks and hence constitute a chain called Blockchain.



5.2 Digital currency

Blockchain: Question and answer 1:

Q 9. Is it possible in Blockchain to remove one or more block from the networks?

Ans 9: Yes, it can be done. There are times when only a specific portion of this online ledger is to be considered. With the help of default options and filters, this can easily be done without making a lot of efforts.

Q 10. What exactly do you know about the security of a block?

Ans 10: Well, a block or the entire blockchain is protected by a strong cryptographic hash algorithm. Each block has a unique hash pointer. Any modification in the block constituents will result in the change in the hash identifier of the block. Therefore, it offers an excellent level of security. Thus, one needs not to worry about the safety as well as the security of data that is present in a block.

Q 11. What are Block Identifiers?

Ans 11: In Blockchain, blocks can be identified by the *block header hash* and the *block height*.

Q 12. Is it possible to modify the data once it is written in a block?

Ans12: No, it's not possible to do so. In case any modification is required, the organization simply has to erase the information from all other blocks too. It is because of no other reason than this, data must be given the extreme care of while using this approach.

Q 10. How does a block is recognized in the Blockchain approach?

Ans 13: Every block in this online ledger basically consists of a hash pointer which acts as a link to the block which is prior to it, transaction data and in fact a stamp of time.

5.3 Cyber security

Learning Objectives:

Candidates should be able to:

- 1 Describe the processes involved in, and the aim of carrying out, a range of cyber security threats

Notes and guidance

- Including:
 - brute-force attack
 - data interception
 - distributed denial of service (DDoS) attack
 - hacking
 - malware (virus, worm, Trojan horse, spyware, adware, ransomware)
 - pharming
 - phishing
 - social engineering

5.3 Cyber security

Learning Objectives:

- 2 Explain how a range of solutions are used to help keep data safe from security threats
 - Including:
 - access levels
 - anti-malware including anti-virus and anti-spyware
 - authentication (username and password, biometrics, two-step verification)
 - automating software updates
 - checking the spelling and tone of communications
 - checking the URL attached to a link
 - firewalls
 - privacy settings
 - proxy-servers
 - secure socket layer (SSL) security protocol

5.3 Cyber security

Cyber security threats:

What Is a Cyber Security Threat?

<https://www.youtube.com/watch?v=w1d81Teltl0>



5.3 Cyber security

Cyber security threats:

What Is a Cyber Security Threat?

A cyber security threat refers to any possible malicious attack that seeks to unlawfully access data, disrupt digital operations or damage information. Cyber threats can originate from various actors, including corporate spies, hacktivists, terrorist groups, hostile nation-states, criminal organizations, lone hackers and disgruntled employees.

In recent years, numerous high-profile cyber-attacks have resulted in sensitive data being exposed. For example, the 2017 Equifax breach compromised the personal data of roughly 143 million consumers, including birth dates, addresses and Social Security numbers. In 2018, Marriott International disclosed that hackers accessed its servers and stole the data of roughly 500 million customers. In both instances, the cyber security threat was enabled by the organization's failure to implement, test and retest technical safeguards, such as encryption, authentication and firewalls.

Cyber attackers can use an individual's or a company's sensitive data to steal information or gain access to their financial accounts, among other potentially damaging actions, which is why cyber security professionals are essential for keeping private data protected.

5.3 Cyber security

Cyber security threats:

Types of Cyber Security Threats

The following list shows the cyber threats which will be considered in this section 1:

- » Brute force attacks
- » Data interception
- » Distributed denial of service (DDoS) attacks
- » Hacking
- » Malware (viruses, worms, Trojan horse, spyware, adware and ransomware)
- » Phishing
- » Pharming
- » Social engineering.

5.3 Cyber security

Cyber security threats:

Brute force attacks:

<https://www.youtube.com/watch?v=9juhDTsmyTI>



Dictionary Attack
Brute Force Attack



5.3 Cyber security

Cyber security threats:

Brute force attacks:

<https://www.youtube.com/watch?v=PIP6i3qkaJk>



5.3 Cyber security

Cyber security threats:

What's a Brute Force Attack?

A brute force attack uses trial-and-error to guess login info, encryption keys, or find a hidden web page. Hackers work through all possible combinations hoping to guess correctly. For example Check if the password is one of the most common ones used (the five most common are: 12345678, password, qwerty, 1234 and abc123 ot Abc1234); since these simple passwords are seen so many times it's a good place for the hacker to start.

These attacks are done by 'brute force' meaning they use excessive forceful attempts to try and 'force' their way into your private account(s).

This is an old attack method, but it's still effective and popular with hackers. Because depending on the length and complexity of the password, cracking it can take anywhere from a few seconds to many years.

What do hackers gain from Brute Force Attacks?

Brute force attackers have to put in a bit of effort to make these schemes pay off. While technology does make it easier, you might still question: why would someone do this?

Here's how hackers benefit from brute force attacks:

Profiting (obtain) from ads or collecting activity data , Stealing personal data and valuables ,Spreading malware to cause disruptions, Hijacking your system for malicious activity and Ruining a website's reputation

5.3 Cyber security

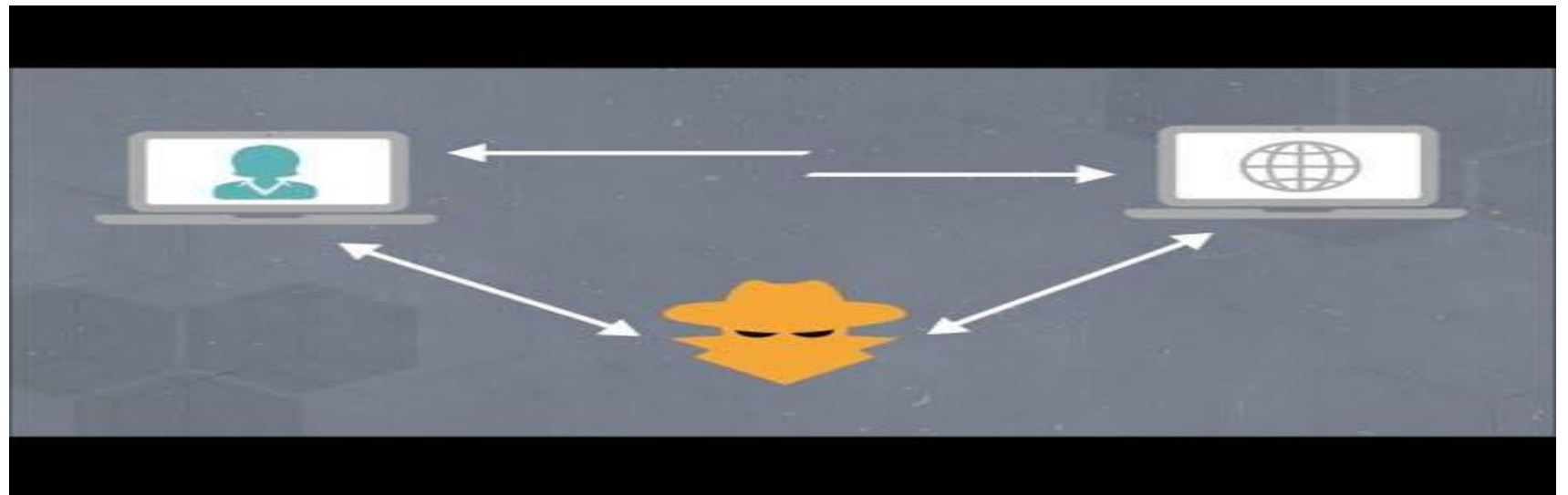
Cyber security threats:

What's a Data interception?

Where data is intercepted during transmission. This is done using software called a packet sniffer (is a technique whereby packet data flowing across the network is detected and observed), which examines data packets as they are sent around a network, or across the internet. The information gathered is sent back to a hacker.

Data interception entails an attacker intercepting a data transfer between a client and a server. The attacker tricks the client and the server into believing that they are exchanging information with each other, while the attacker intercepts the data, creates a connection to the real site and acts as a proxy to read and insert false information into the communication.

https://youtu.be/83LOa-dYi_A



5.3 Cyber security

Cyber security threats:

Denial of service attacks:

Watch the video for Denial of service attacks (DoS)

<https://www.youtube.com/watch?v=ilhGh9CElwM>



5.3 Cyber security

Cyber security threats:

Denial of service attacks:

(DoS): is an attempt at preventing users from accessing part of a network, notably an internet server. This is usually temporary but may be a very damaging act or a big breach of security. It doesn't just affect networks; an individual can also be a target for such an attack. One method of attack is to flood the network with useless traffic.

The attacker may be able to prevent a user from:

- accessing their emails
- accessing websites/web pages
- accessing online services (such as banking).

How does this cause the problem?

- When a user types in or clicks on a URL of a website (using their web browser), a request is sent to the internet server which contains the website or web page. Obviously, the server can only handle a finite number of requests. So if it becomes overloaded by an attacker sending out thousands of requests, it won't be able to service the user's legitimate request. This is effectively a denial of service.
- This can happen to a user's email account, for example, by an attacker sending out many spam messages to their email account. Internet Service Providers (ISPs) only allow a specific data quota for each user. Consequently, if the attacker sends out very large or many hundreds of emails to the user's account, it will quickly get clogged up and the user won't be able to receive legitimate emails.

5.3 Cyber security

Cyber security threats:

Denial of service attacks:

How to guard against these attacks:

- using an up-to-date malware/virus checker
- setting up a firewall to restrict traffic
- applying email filters to manage or filter out unwanted traffic or spam emails.

5.3 Cyber security

Hacking

Watch the video for difference between hacker

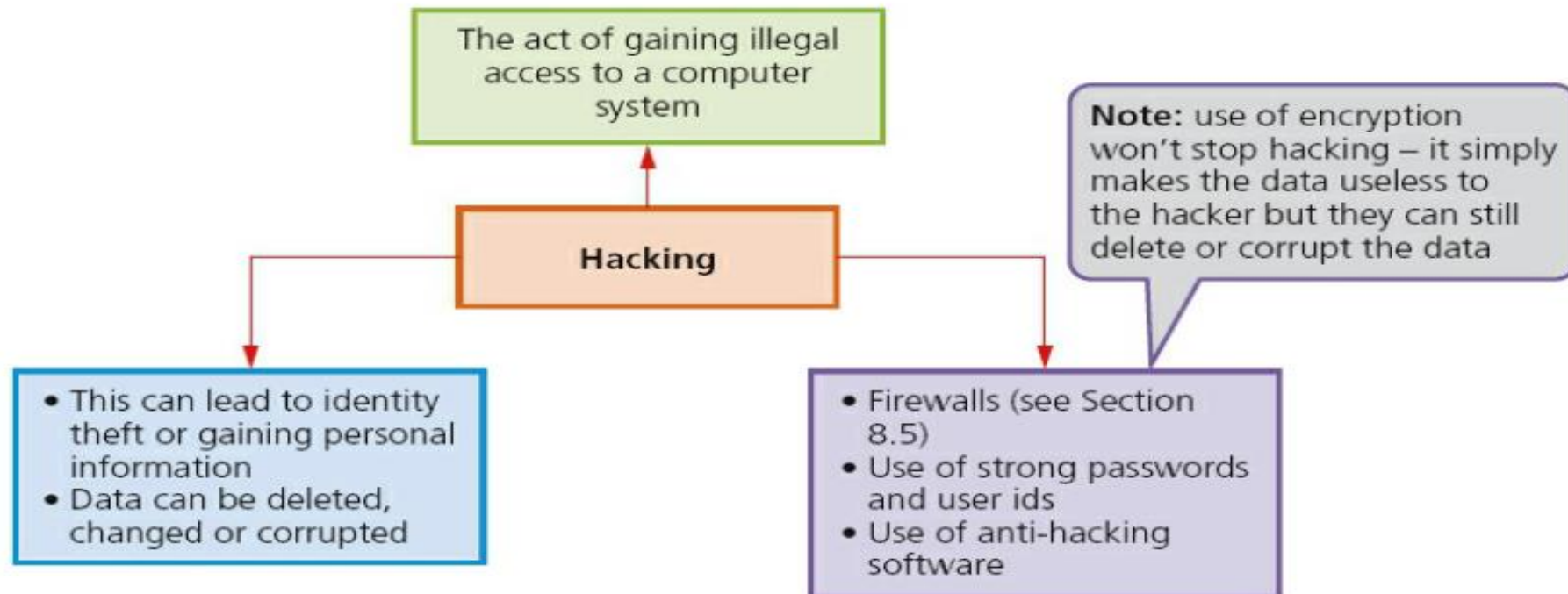
https://www.youtube.com/watch?v=h5_Q7dbaJDE



5.3 Cyber security

Hacking and Cracking

Hacking is breaking into a computer system to steal personal data without the owner's consent or knowledge (e.g. to steal a password file).



5.3 Cyber security

Note the difference between **CRACKING** and **HACKING**.

- **Cracking**: is where someone edits a program source code (i.e. looks for a 'back door' in the software so that the code can be exploited or changed for a specific purpose). This is usually done for a malicious purpose (e.g. legitimate software could be altered by a cracker to perform a different task e.g. send a user to a specific website).
- **Hacking**: is breaking into a computer system to steal personal data without the owner's consent or knowledge (e.g. to steal a password file).

Essentially, hacking isn't necessarily harmful whilst cracking is ALWAYS totally illegal and is potentially very damaging.

5.3 Cyber security

Watch the video for **Malware**: Difference Between Computer Viruses, Worms and Trojans

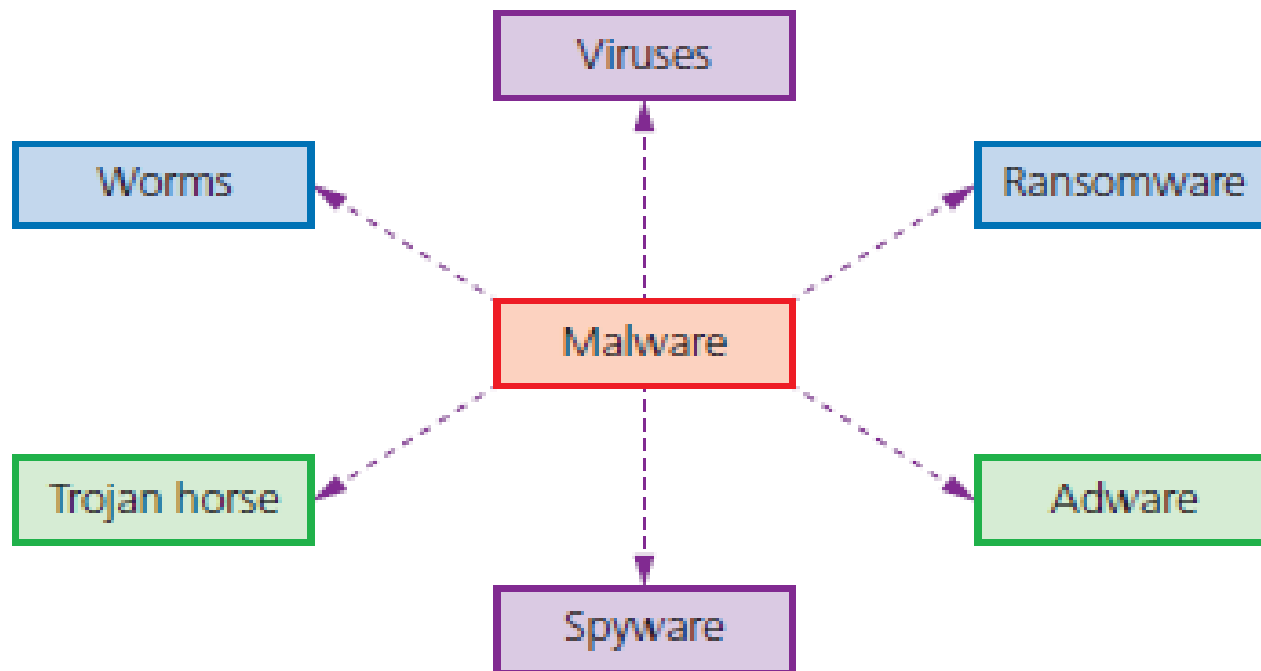
<https://www.youtube.com/watch?v=n8mbzU0X2nQ>



5.3 Cyber security

Malware:

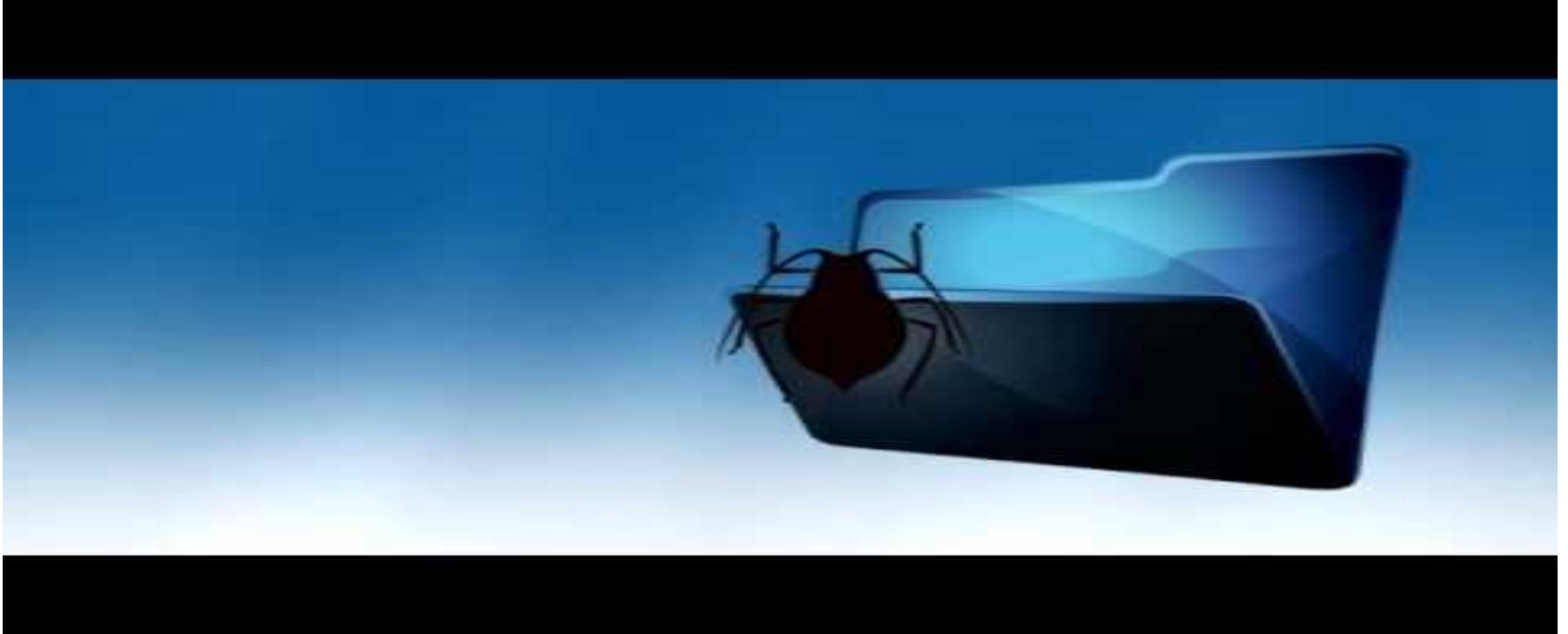
Malware is one of the biggest risks to the integrity and security of data on a computer system. There are many forms of malware; this chapter will only consider the following in any detail:



5.3 Cyber security

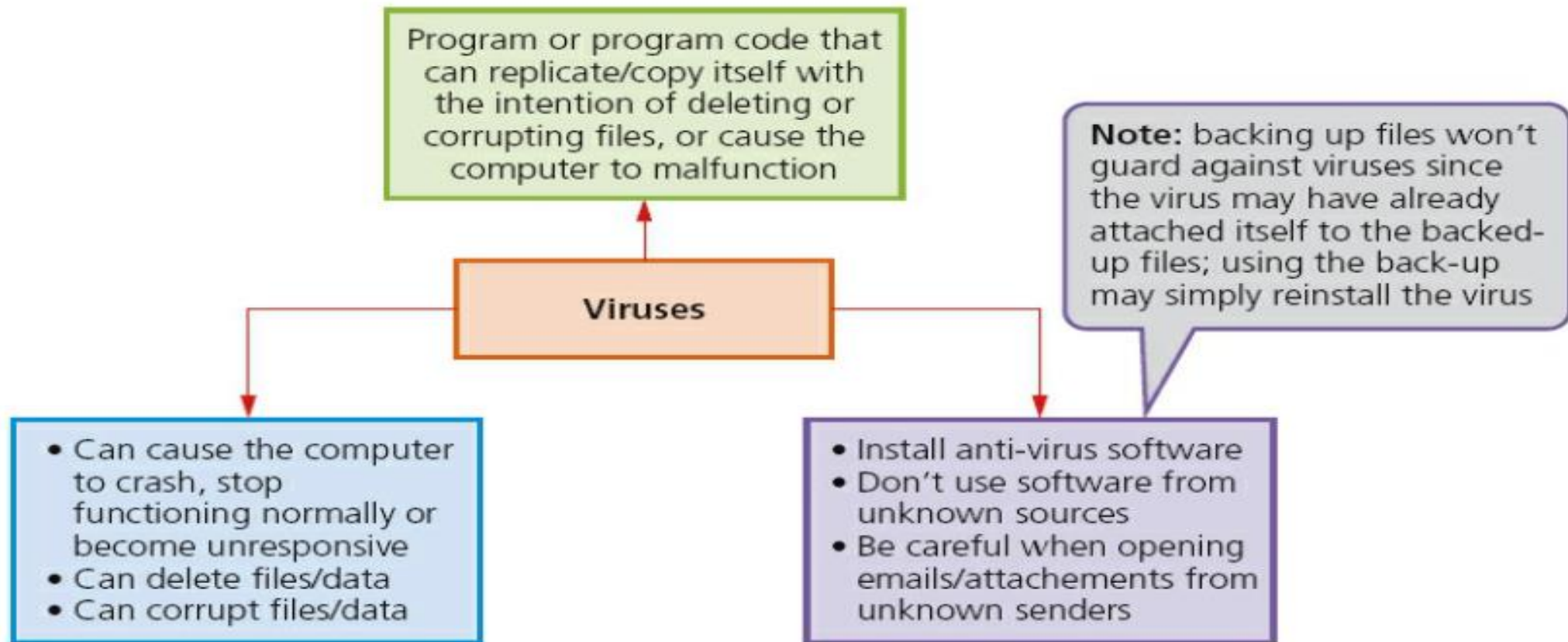
Viruses

<https://youtu.be/qNodotqsUgg>



5.3 Cyber security

A virus is a piece of code that replicates itself and causes damage to computer systems by inserting their own code.



5.3 Cyber security

What is the difference between antivirus and anti-malware?

https://youtu.be/fKxuKWsA_JI

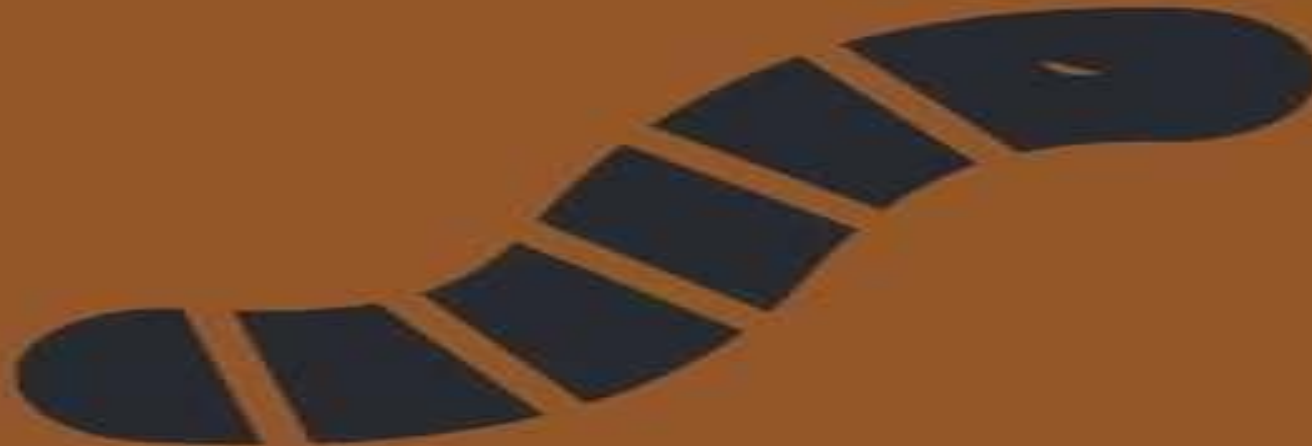


5.3 Cyber security

Worm

A computer worm is **a subset of the Trojan horse malware that can propagate or self-replicate from one computer to another without human activation after breaching a system.** Typically, a worm spreads across a network through your Internet or LAN (Local Area Network) connection.

<https://www.youtube.com/watch?v=oyUsZu6ygq8>



5.3 Cyber security

Trojan Horse

<https://www.youtube.com/watch?v=pzOM8sc2RPU>



5.3 Cyber security

Trojan Horse

A Trojan Horse Virus is **a type of malware that downloads onto a computer disguised as a legitimate program**. The delivery method typically sees an attacker use social engineering to hide malicious code within legitimate software to try and gain users' system access with their software.

Here is one example of how a Trojan horse might be used to infect a personal computer: **The victim receives an official-looking email with an attachment. The attachment contains malicious code that is executed as soon as the victim clicks on the attachment.**

How to protect against a Trojan horse

The easiest way to protect a system from a Trojan horse is by never opening or downloading emails or attachments from unknown sources. Deleting these messages before opening will delete the Trojan horse threat.

5.3 Cyber security

Spyware

Spyware is software with malicious behavior that aims to gather information about a person or organization and send it to another entity in a way that harms the user—for example, by violating their privacy or endangering their device's security.

<https://www.youtube.com/watch?v=-Z3pp14oUiA>



5.3 Cyber security

Adware

Adware is software that displays unwanted (and sometimes irritating) pop-up adverts which can appear on your computer or mobile device.

<https://www.youtube.com/watch?v=jlylqjOlqzY>



5.3 Cyber security

Ransomware

Ransomware is a malware designed to **deny a user or organization access to files on their computer**. By encrypting these files and demanding a ransom payment for the decryption key, cyber attackers place organizations in a position where paying the ransom is the easiest and cheapest way to regain access to their files.

<https://www.youtube.com/watch?v=Vkjekr6jacg>



5.3 Cyber security

Practical Encryption 2: Open Python IDLE and Enter the code below:

The *fernet* module of the *cryptography* package has inbuilt functions for the generation of the key, encryption of plain text into cipher text, and decryption of cipher text into plain text using the *encrypt()* and *decrypt()* methods respectively.

The *fernet* module guarantees that data encrypted using it cannot be further manipulated or read without the key.

We are going to use the [Movies.csv](#) file to perform all operations

In the cryptography library, there is a cryptography algorithm called fernet.

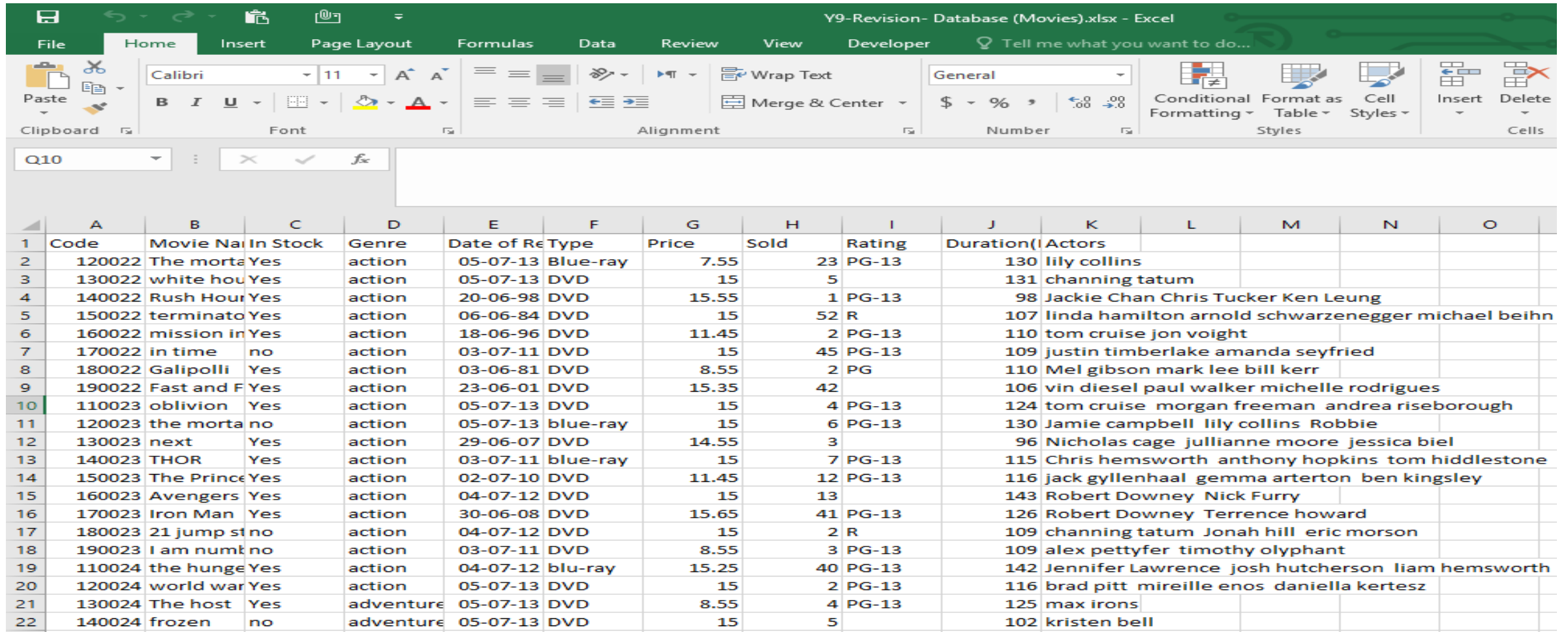
We will use the fernet module to encrypt the file.

Fernet guarantees that a message encrypted using it cannot be manipulated or read without the key. **Fernet** is an implementation of symmetric (also known as “secret key”) authenticated cryptography

5.3 Cyber security

Practical Encryption 2: Open Python IDLE and Enter the code below:

[Movies.csv](#) file to perform all operations to **encrypt()**



The screenshot shows the Microsoft Excel interface with the 'Home' tab selected. The spreadsheet contains a movie database with columns for Code, Movie Name, In Stock, Genre, Date of Release, Type, Price, Sold, Rating, Duration, and Actors. The data is organized into rows, with some cells containing multiple actor names separated by spaces.

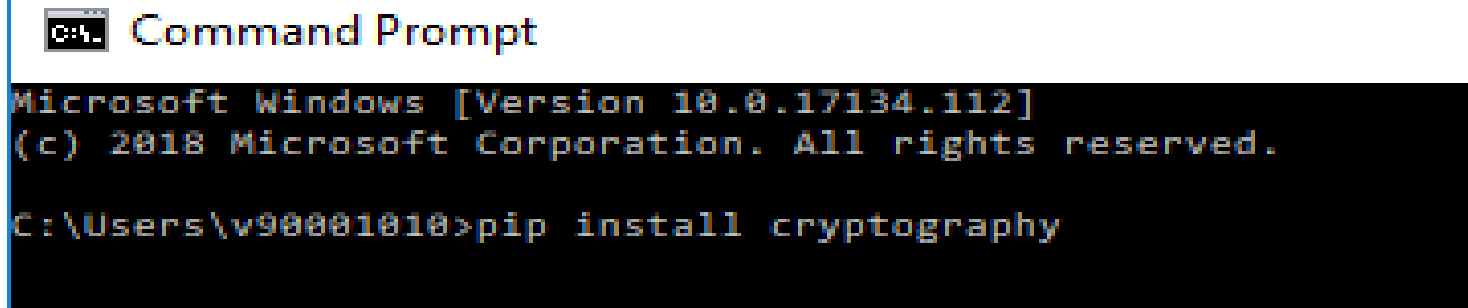
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	Code	Movie Name	In Stock	Genre	Date of Release	Type	Price	Sold	Rating	Duration	Actors				
1															
2	120022	The mortal	Yes	action	05-07-13	Blue-ray	7.55	23	PG-13	130	lily collins				
3	130022	white hou	Yes	action	05-07-13	DVD	15	5		131	channing tatum				
4	140022	Rush Hour	Yes	action	20-06-98	DVD	15.55	1	PG-13	98	Jackie Chan Chris Tucker Ken Leung				
5	150022	terminato	Yes	action	06-06-84	DVD	15	52	R	107	linda hamilton arnold schwarzenegger michael beihn				
6	160022	mission in	Yes	action	18-06-96	DVD	11.45	2	PG-13	110	tom cruise jon voight				
7	170022	in time	no	action	03-07-11	DVD	15	45	PG-13	109	justin timberlake amanda seyfried				
8	180022	Galipolli	Yes	action	03-06-81	DVD	8.55	2	PG	110	Mel gibson mark lee bill kerr				
9	190022	Fast and F	Yes	action	23-06-01	DVD	15.35	42		106	vin diesel paul walker michelle rodrigues				
10	110023	oblivion	Yes	action	05-07-13	DVD	15	4	PG-13	124	tom cruise morgan freeman andrea riseborough				
11	120023	the morta	no	action	05-07-13	blue-ray	15	6	PG-13	130	Jamie campbell lily collins Robbie				
12	130023	next	Yes	action	29-06-07	DVD	14.55	3		96	Nicholas cage jullianne moore jessica biel				
13	140023	THOR	Yes	action	03-07-11	blue-ray	15	7	PG-13	115	Chris hemsworth anthony hopkins tom hiddlestone				
14	150023	The Prince	Yes	action	02-07-10	DVD	11.45	12	PG-13	116	jack gyllenhaal gemma arterton ben kingsley				
15	160023	Avengers	Yes	action	04-07-12	DVD	15	13		143	Robert Downey Nick Furry				
16	170023	Iron Man	Yes	action	30-06-08	DVD	15.65	41	PG-13	126	Robert Downey Terrence howard				
17	180023	21 jump st	no	action	04-07-12	DVD	15	2	R	109	channing tatum Jonah hill eric morson				
18	190023	I am num	no	action	03-07-11	DVD	8.55	3	PG-13	109	alex pettyfer timothy olyphant				
19	110024	the hunge	Yes	action	04-07-12	blu-ray	15.25	40	PG-13	142	Jennifer Lawrence josh hutcherson liam hemsworth				
20	120024	world war	Yes	action	05-07-13	DVD	15	2	PG-13	116	brad pitt mireille enos daniella kertes				
21	130024	The host	Yes	adventure	05-07-13	DVD	8.55	4	PG-13	125	max irons				
22	140024	frozen	no	adventure	05-07-13	DVD	15	5		102	kristen bell				

5.2 Digital currency:

Practical Encryption 2:

Building cryptography on Windows: *encrypt()*

Open **cmd** and type **pip install cryptography** and press enter as shown below:



```
C:\> Command Prompt

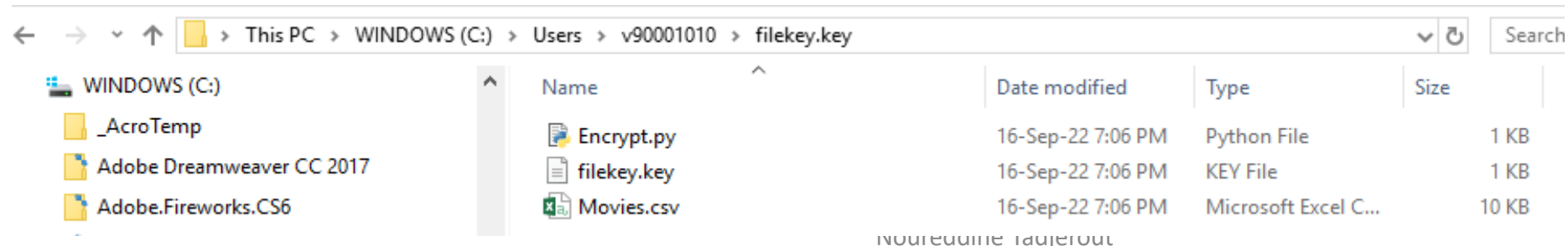
Microsoft Windows [Version 10.0.17134.112]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\v90001010>pip install cryptography
```

Next create a folder on the c drive as shown below:

C:\Users\v90001010\filekey.key

Add a Movies.csv file as shown



5.3 Cyber security

Practical Encryption 2: Open Python IDLE and Enter the code below:

```
*Encrypt.py - C:/Users/v90001010/filekey.key/Encrypt.py (3.10.6)*
File Edit Format Run Options Window Help
# import required module
from cryptography.fernet import Fernet
# key generation
key = Fernet.generate_key()

# string the key in a file
with open('filekey.key', 'wb') as filekey:
    filekey.write(key)

# opening the key
with open('filekey.key', 'rb') as filekey:
    key = filekey.read()

# using the generated key
fernet = Fernet(key)

# opening the original file to encrypt
with open('Movies.csv', 'rb') as file:
    original = file.read()

# encrypting the file
encrypted = fernet.encrypt(original)

# opening the file in write mode and
# writing the encrypted data
with open('Movies.csv', 'wb') as encrypted_file:
    encrypted_file.write(encrypted)
```

In the cryptography library, there is a cryptography algorithm called fernet.

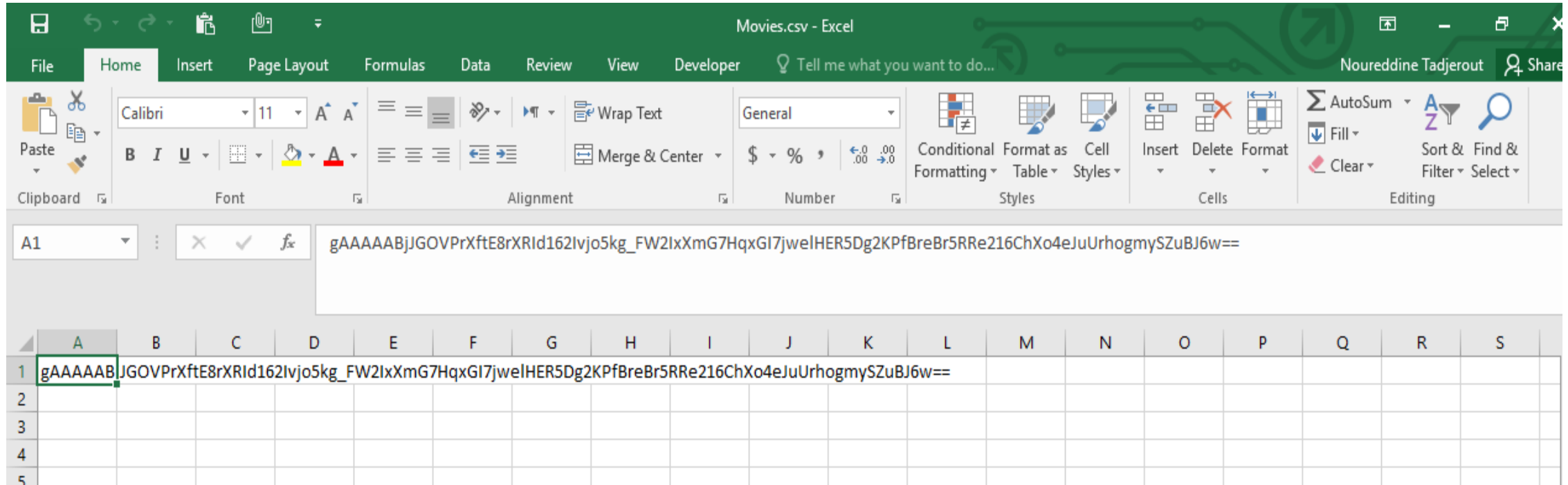
We will use the fernet module to encrypt the file.

Fernet guarantees that a message encrypted using it cannot be manipulated or read without the key. Fernet is an implementation of symmetric (also known as “secret key”) authenticated cryptography

5.3 Cyber security

Practical Encryption 2: Open Python IDLE and Enter the code below:

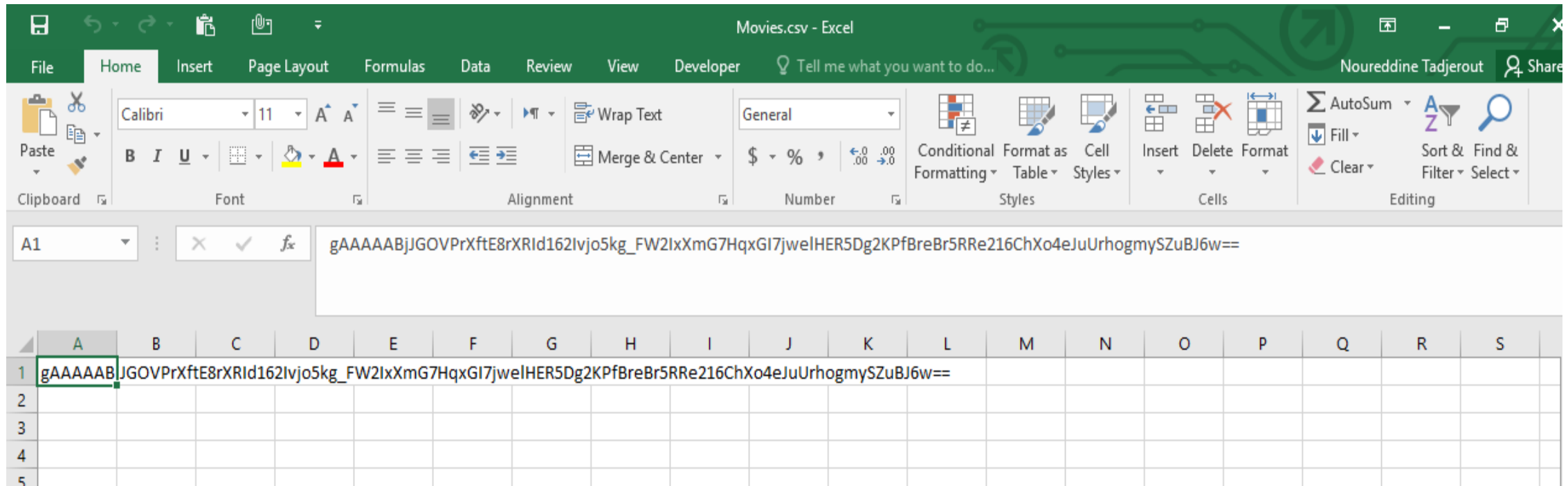
Below the Encrypted Movie.csv



5.3 Cyber security

Practical Encryption 2: Open Python IDLE and Enter the code below:

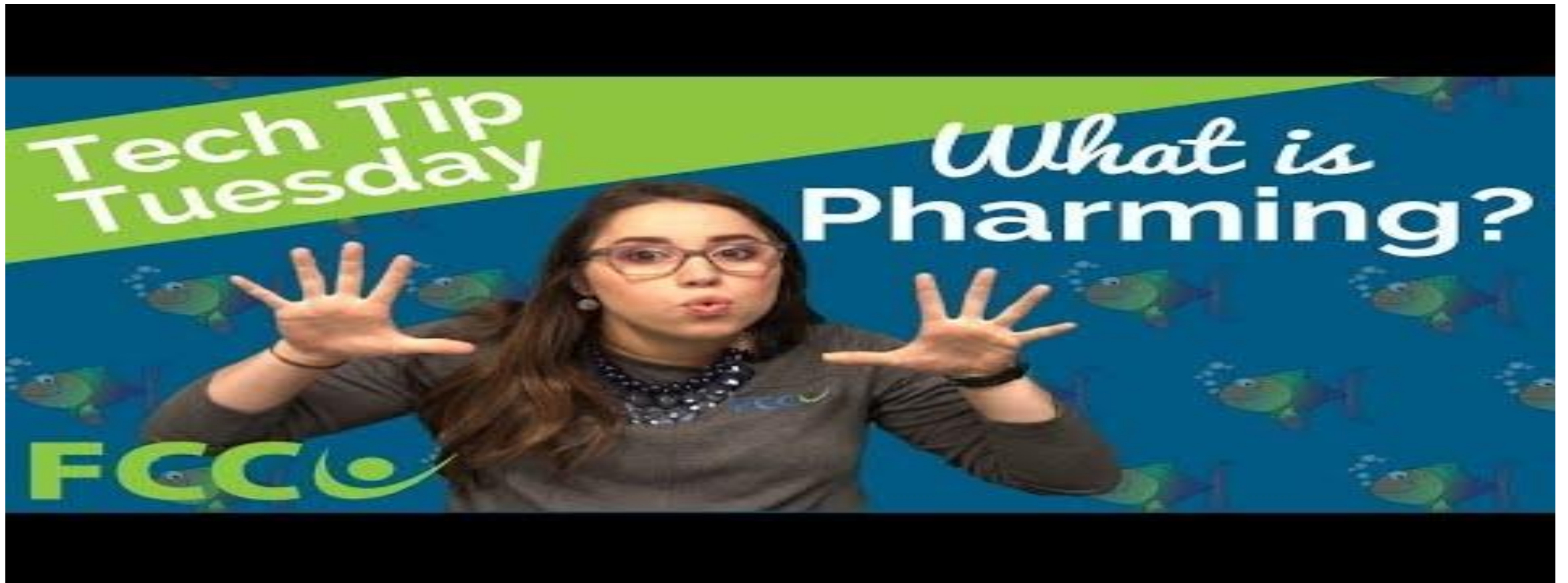
Now your task is write python program to decrypt movie.csv below



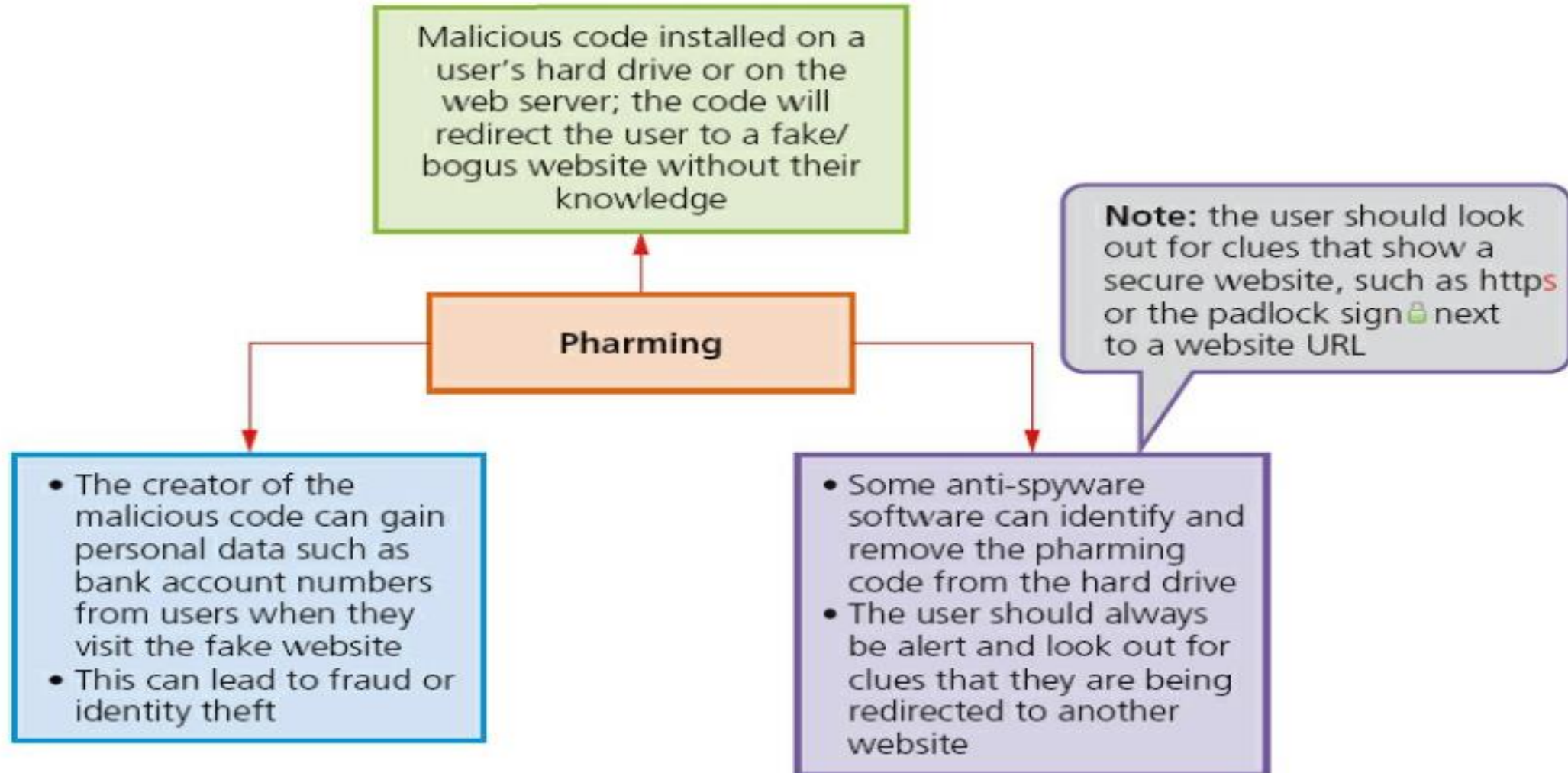
5.3 Cyber security

Pharming:

<https://www.youtube.com/watch?v=4F89EvGJ2wA>



5.3 Cyber security



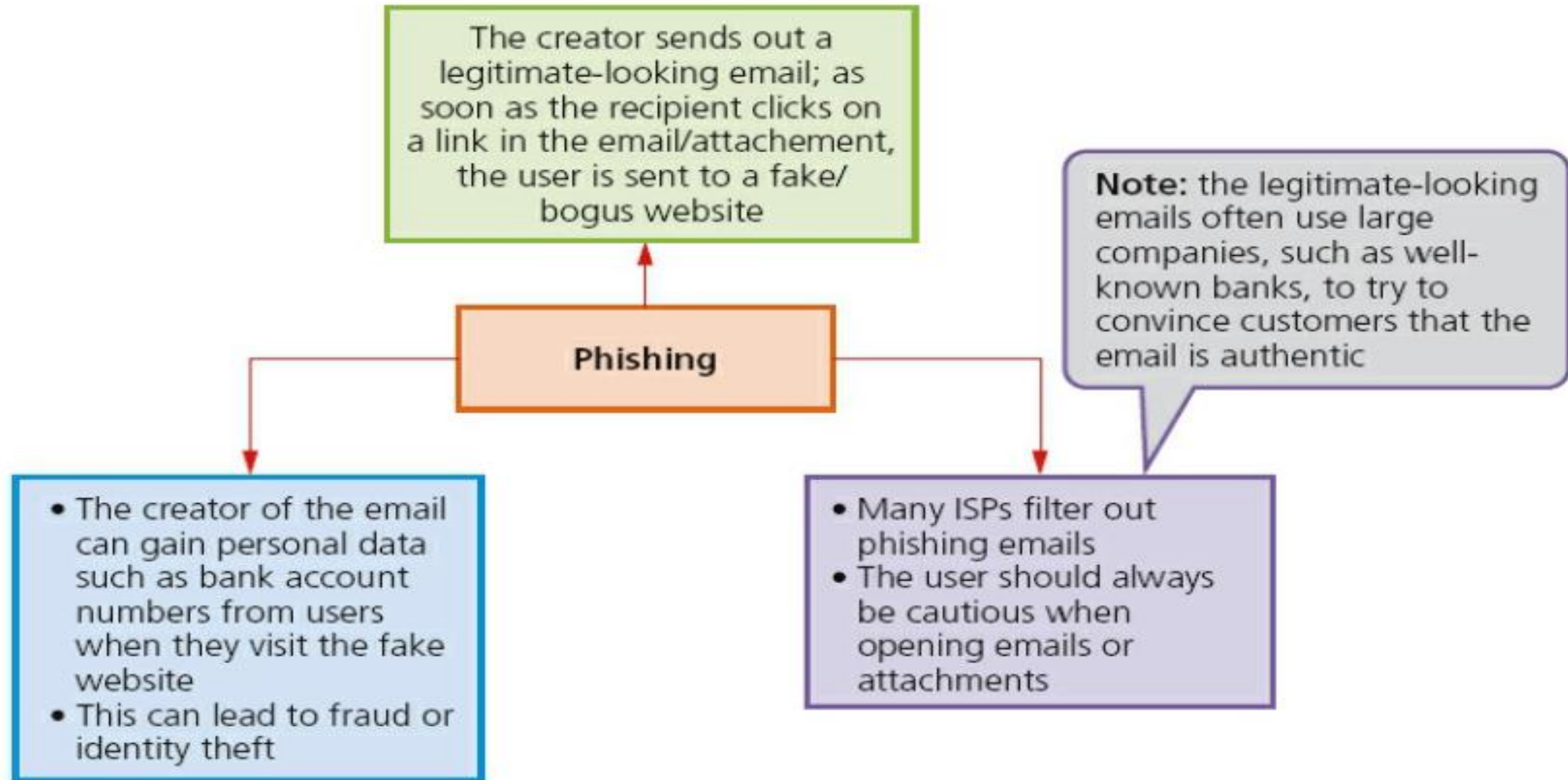
5.3 Cyber security

Phishing

<https://www.youtube.com/watch?v=Y7zNIEMDmI4>



5.3 Cyber security



5.3 Cyber security

Phishing



5.3 Cyber security

Social Engineering

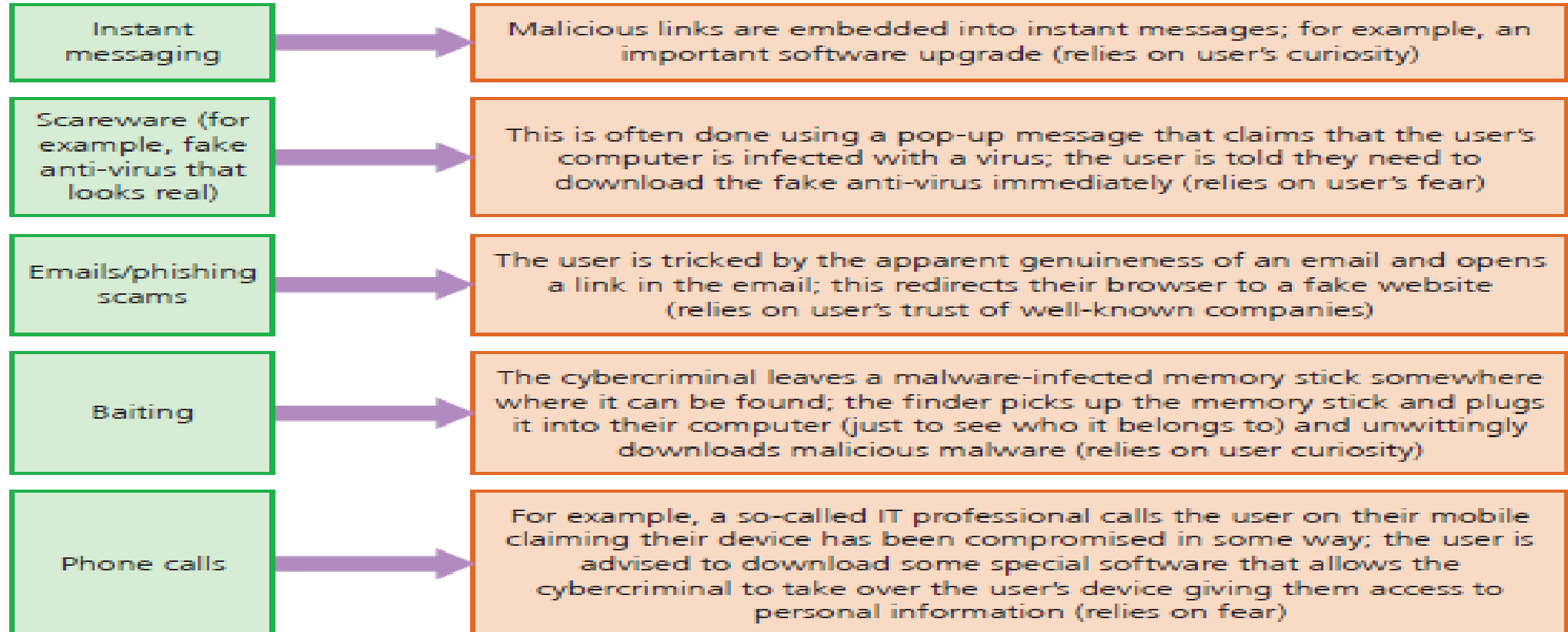
<https://www.youtube.com/watch?v=Vo1urF6S4u0>



5.3 Cyber security

Social Engineering

Social engineering is the act of exploiting human weaknesses to gain access to personal information and protected systems. Social engineering relies on manipulating individuals rather than hacking computer systems to penetrate a target's account. An example is an email sent to users of an online service that alerts them of a policy violation requiring immediate action on their part, such as a required password change.



5.3 Cyber security

Learning Objectives

- 2 Explain how a range of solutions are used to help keep data safe from security threats
 - Including:
 - access levels
 - anti-malware including anti-virus and anti-spyware
 - authentication (username and password, biometrics, two-step verification)
 - automating software updates
 - checking the spelling and tone of communications
 - checking the URL attached to a link
 - firewalls
 - privacy settings
 - proxy-servers
 - secure socket layer (SSL) security protocol

5.3 Cyber security

Keeping data safe from security threats

Access Level:

Refers to a type of Manager permission that provides more control over what data they can access without having to set multiple permissions.

Access levels are **permission sets that allow members to perform different tasks within an organization**. By assigning organization members to access levels, you determine the information they can see and actions they are allowed to perform. Important: Access levels are not system roles.

The components of an access level include:

Administrator: Full access to the system. Allows the user to perform most operations on the system. Can add, remove, and edit users, groups, and domains; manage passwords; configure services; and more.

Standard: Limited access to the system. Does not allow the user to perform many functions. Cannot delete, rename, move, copy, or format drives. Only allows the user to print documents.

Guest: No access to the system. Users with guest accounts can log onto the computer, but they cannot perform any actions. They cannot use the mouse, open windows, or even save their work.



5.3 Cyber security

Keeping data safe from security threats

Anti-Malware including anti-virus and Anti-Spyware:

How Does Antivirus and Antimalware Software Work?

<https://youtu.be/bTU1jbVXlmM>



5.3 Cyber security

Keeping data safe from security threats

Anti-Malware including anti-virus and Anti-Spyware:

What Does Antivirus mean?

Antivirus software scans your device for known viruses and searches for any known threats. They monitor the behaviour of all programs on a system and notify of any suspicious behaviours. The goal of antivirus software is to block and remove all malware found as quickly as possible.

It is important to have antivirus software installed, especially with the constant emerging of cyber threats. Therefore, if you do not have antivirus protection, you “could be at risk of picking up a virus or benign targeted by other malicious software that can remain undetected and wreak havoc on your computer and mobile devices” (reference from **Norton**).

The following characteristics indicate good antivirus software:

Real-time scanning: the software is continuously scanning your system and detect/react to threats as they encounter them

Automatic updates: automatically updates your system once malware has been found.

Remove threats: the software removes the malware that has been found on your system

5.3 Cyber security

Keeping data safe from security threats

Anti-Malware including anti-virus and Anti-Spyware:

What Does Anti-malware mean?

Anti-malware software “defends against second-generation malware that classic antivirus software doesn’t always detect”. It protects “users from the latest, currently in the wild, and even more dangerous threats”. It has the capability to update its rules faster than antivirus software can, which means that it is the best protection against new malware you may encounter while browsing the web.

Anti-malware uses three different techniques for malware detection: signature-based, behavior-based, and sandboxing.

Signature-Based Detection

This type of detection uses “a set of known software components and their digital signatures to identify new malicious software”. These signatures are then used to “identify previously identified malicious software of the same type and to flag the new software as malware.”

Behaviour-Based Detection

This detection works by “identifying malicious software by examining how it behaves rather than what it looks like”.

5.3 Cyber security

Keeping data safe from security threats

Anti-Malware including anti-virus and Anti-Spyware:

What Does Anti-malware mean?

Sandboxing

Sandboxing is an isolated environment from the rest of the system used to test potentially malicious files. It is “used as a method to filter out potentially malicious files and remove them before they have had a chance to do damage”.

These are the following characteristics that you should pay attention to when searching for the appropriate anti-malware software:

- Sandboxing
- Traffic filtering: with traffic filtering, the software should protect your device by blocking access to suspicious servers and sites involved with malware distribution
- Proactive security: the software should scan, detect and remove known malware threats such as spyware, trojans, and adware.

The main difference between antivirus and anti-malware is that antivirus software “can protect against common types of viruses while anti-malware software works to detect new iterations of infections”

5.3 Cyber security

Keeping data safe from security threats

Anti-Malware including anti-virus and Anti-Spyware:

How to Protect Your Computer from Spyware:

The best way to control spyware is by preventing it from getting on your computer in the first place. However, avoiding program downloads and email attachments isn't always an option. Sometimes, even a trusted website can become compromised and infect your computer.

- Stay away from unofficial app stores
- Only download trusted apps from official app publishers.
- Be reserved about giving permissions to apps.
- Do not follow links in text messages.
- Enable or download a pop-up blocker.
- Limit runnable applications to a pre-approved allow list.
- Avoid email links and attachments when possible.
- blocks access to a user's webcam and microphone.

5.3 Cyber security

Keeping data safe from security threats

Authentication (username and password, biometrics, two-step verification)

Authentication:

Authentication is **the process of determining whether someone or something is, in fact, who or what it says it is**. Authentication technology provides access control for systems by checking to see if a user's credentials match the credentials in a database of authorized users or in a data authentication server.

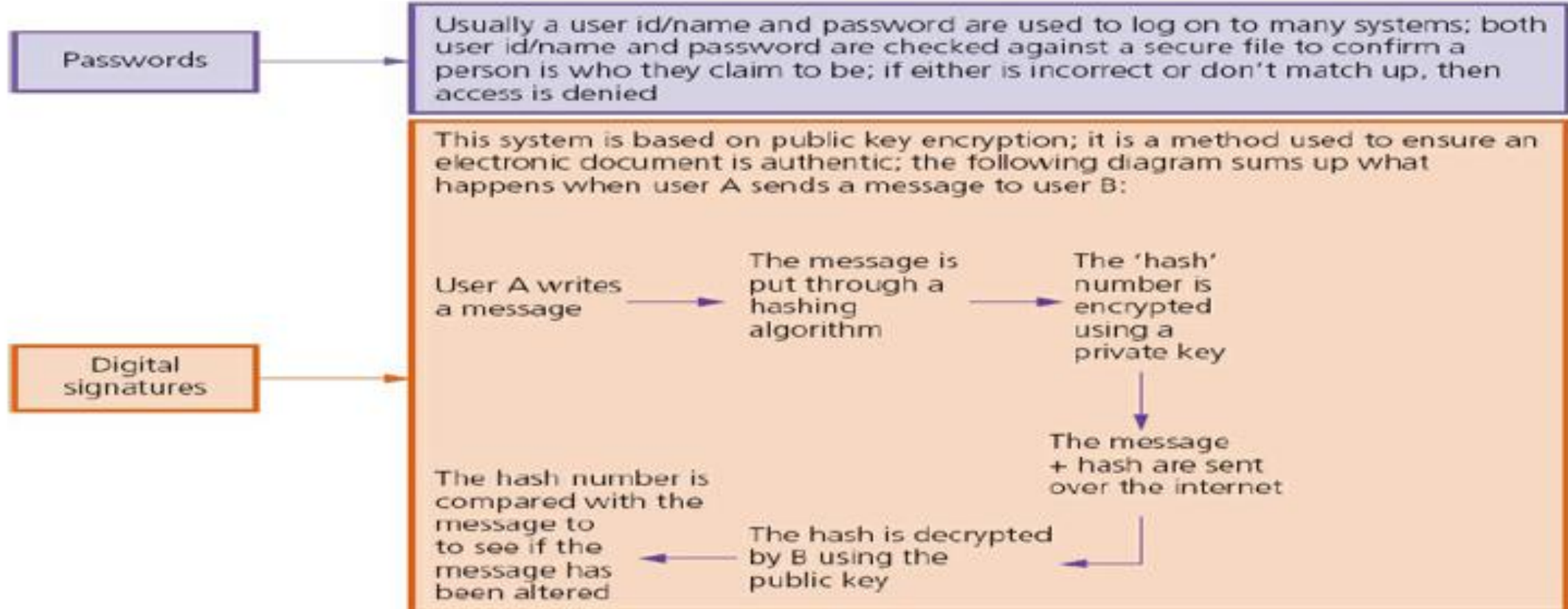
Example: Username and password combination is the most popular authentication mechanism, and it is also known as password authentication. A well-known example is **accessing a user account on a website or a service provider such as Facebook or Gmail**

5.3 Cyber security

Keeping data safe from security threats

Authentication (username and password, biometrics, two-step verification)

Authentication: is used to verify that data comes from a trusted source. It works with encryption to strengthen internet security. For example:



5.3 Cyber security

Keeping data safe from security threats

Authentication (username and password, biometrics, two-step verification)

Passwords and User names:

A password is a string of characters used **to verify the identity of a user during the authentication process**. Passwords are typically used in tandem (mean “**having two things arranged one in front of the other**”) with a username; they are designed to be known only to the user and allow that user to gain access to a device, application or website.

Passwords can vary in length and can contain letters, numbers and special characters.

How to create a secure password

Passwords, when carefully created and protected, increase safe and secure interactions online and in the workplace and can prevent [password cracking](#). To maximize the strength and efficacy of passwords, organizations often establish password policies. These policies are designed to help users create [strong passwords](#) and adopt best practices for managing login credentials. Below are a few examples of the practices that contribute to effective password management and creation:

A minimum length of eight characters and a maximum between 16 to 64 characters. Include both uppercase and lowercase letters with case sensitivity.

Use at least one number and Use at least one special character. Very good password: K7&\$IHio66Nay

Avoid using easily guessed elements such as names of children, pet names and birthdays.

5.3 Cyber security

Keeping data safe from security threats

Authentication (username and password, biometrics, two-step verification)

Biometrics:

Biometric methods authenticates users based on physiological characteristics such as fingerprints or retinal scans or behavioral characteristics such as typing patterns and voice recognition.

Biometrics scanners are hardware used to capture the biometric for verification of identity. **These scans match against the saved database to approve or deny access to the system.** In other words, biometric security means your body becomes the “key” to unlock your access.

Here are some common examples of biometric security:

- Facial Recognition
- Fingerprint Scanning
- Retina scans
- Voice Recognition
- Heart-Rate Sensors

5.3 Cyber security

Keeping data safe from security threats

Authentication (username and password, biometrics, two-step verification)

Biometrics: Facial Recognition

A facial recognition system is a technology capable of identifying or verifying a person from a digital image or a video frame from a video source. There are multiple methods in which facial recognition systems work, but in general, they work by comparing selected facial features from given image with faces within a database. It is also described as a Biometric Artificial Intelligence based application that can uniquely identify a person by analyzing patterns based on the person's facial textures and shape.

How does Facial Recognition work?

Technologies vary, but here are the basic steps:

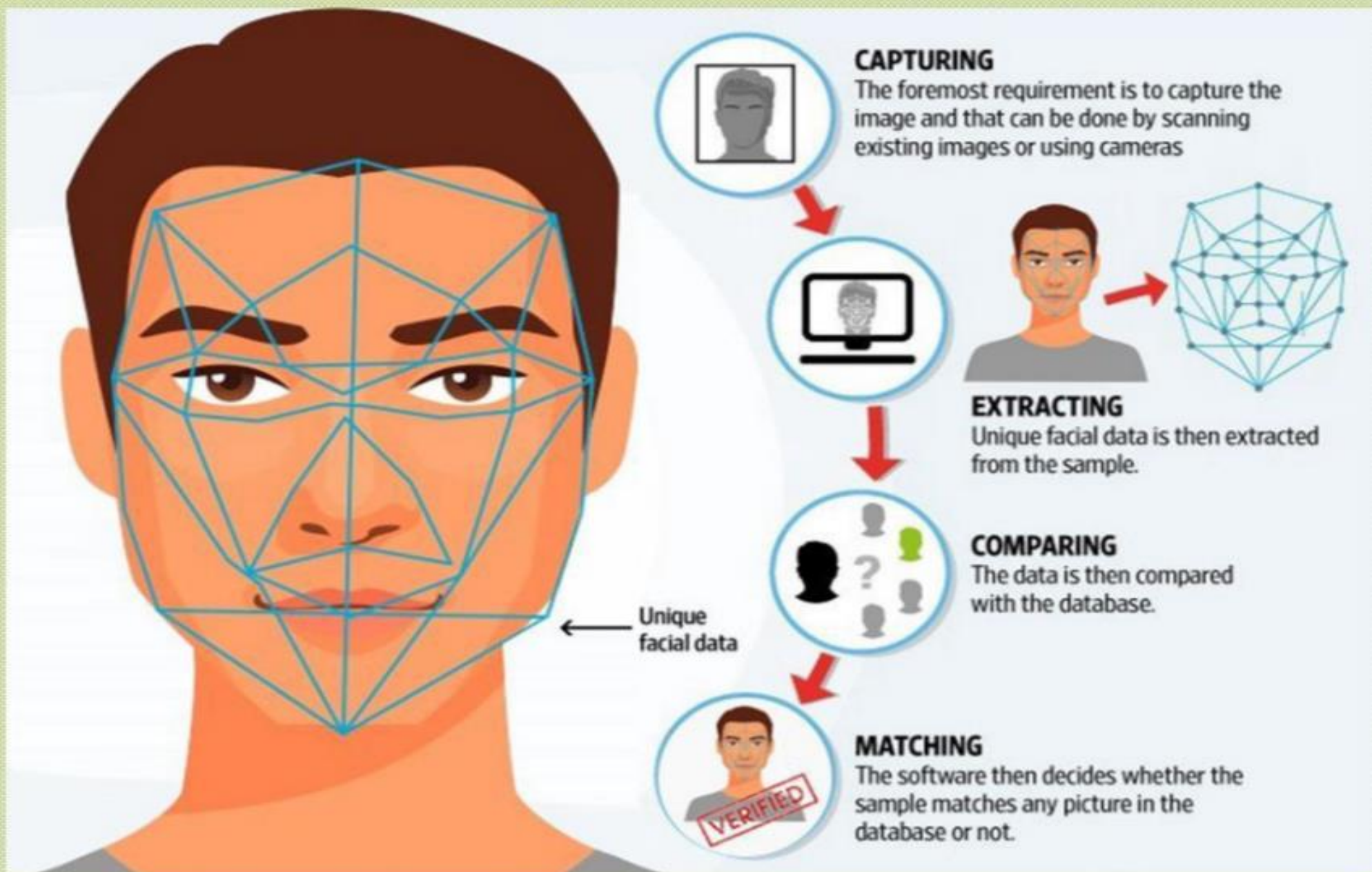
Step 1. A picture of your face is captured from a photo or video.

Step 2. Facial recognition software reads the geometry of your face.

Step 3. Your facial signature — a mathematical formula — is compared to a database of known faces.

Step 4. A determination is made. Your face print may match that of an image in a facial recognition system database.

How Face Recognition Software Works



SOME OF THE PLACES SUCH SYSTEMS ARE EITHER BEING USED OR COULD BE USED



Airports



Railway stations



Banks & financial institutions



Stadiums



Public transport



Government offices



Business establishments

5.3 Cyber security

Keeping data safe from security threats

Authentication (username and password, biometrics, two-step verification)

Biometrics: Retina scans

it is a rather unpleasant technique requiring a person to sit totally still for 10 to 15 seconds while the scan takes place; it is very secure since nobody has yet found a way to duplicate the blood vessels patterns. The accuracy is about 1 in 10 million. Retina scanning is a biometric verification technology that uses an image of an individual's retinal blood vessel pattern as a unique and retina scans use infrared light to scan the unique pattern of blood vessels in the retina (at the back of the eye).

Retina scanners are in use in many military bases, nuclear reactors and other high-security locations due to their strength as a security measure. Retina scans are nearly impossible to fake. Furthermore, because the retina decays so quickly after death, a scan can only be accessed from a living human.



5.3 Cyber security

Keeping data safe from security threats

Authentication (username and password, biometrics, two-step verification)

Biometrics:

Biometric technique	Benefits	Drawbacks
fingerprint scans	it is one of the most developed biometric techniques very easy to use relatively small storage requirements for the biometric data created	for some people it is very intrusive, since it is still related to criminal identification it can make mistakes if the skin is dirty or damaged (e.g. cuts)
retina scans	very high accuracy there is no known way to replicate a person's retina	it is very intrusive it can be relatively slow to verify retina scan with stored scans very expensive to install and set up
face recognition	non-intrusive method relatively inexpensive technology	it can be affected by changes in lighting, the person's hair, change in age, and if the person is wearing glasses
voice recognition	non-intrusive method verification takes less than 5 seconds relatively inexpensive technology	a person's voice can be easily recorded and used for unauthorised access low accuracy an illness such as a cold can change a person's voice, making absolute identification difficult or impossible

5.3 Cyber security

Keeping data safe from security threats

Authentication (username and password, biometrics, two-step verification)

Two-Step Verification:

2FA requires users to provide two authentication factors that include a combination of something the user knows (like a password or PIN), something the user has (like an ID card, security token or smartphone) and something the user is (like a fingerprint or eye scan).

<https://www.youtube.com/watch?v=H-DiXjk8-0w>



5.3 Cyber security

Keeping data safe from security threats

Automating Software Updates

Automatic updates **allow users to keep their software programs updated without having to check for and install available updates manually**. The software automatically checks for available updates, and if found, the updates are downloaded and installed without user intervention.

Checking the spelling and tone of communications

Malicious emails can be difficult to identify, but there are some steps you should take to determine whether emails are legitimate or fake:

Check that the sender's email address has a valid username and domain name. A suspicious email address could be similar to the one below:

“nouredine.tadjerout < noureddinetadjerout.%r@gkmail.com >”.

Verify that you know the sender of an email and that its tone is consistent with the sender.

Look for **grammatical errors or typos** in the body of the message. Companies want to maintain a high degree of professionalism and generally do not send out emails that contain these types of errors.

Consider the tone of the email or what is being offered. If the email is threatening or sounds too good to be true, then it is probably a phishing email.

Pay attention to what is being requested. Most companies do not ask for sensitive or personal information in an email.

5.3 Cyber security

Keeping data safe from security threats

Here are some practical steps you can take today to tighten up your data security.

1. Back up your data.
2. Use strong passwords.
3. Take care when working remotely.
4. Be wary of suspicious emails.
5. Install anti-virus and malware protection.
6. Don't leave paperwork or laptops unattended.
7. Make sure your Wi-Fi is secure,

5.3 Cyber security

Keeping data safe from security threats

Checking the URL attached to a link

How do you know if a link is a phishing link?

There are two parts to a link: The words describing the link (the part you see) and the URL. If you're on a computer, **hover over the link and find out its real destination. If it doesn't match the link displayed, assume it's unsafe and don't click it**

5.3 Cyber security

Keeping data safe from security threats

Firewalls

<https://www.youtube.com/watch?v=kDEX1HXybrU>



5.3 Cyber security

Keeping data safe from security threats

Firewalls

Can be either software or hardware. It sits between the user's computer and an external network (e.g. the internet) and filters information in and out of the Tasks carried out by a firewall include:

examining the 'traffic' between the user's computer (or internal network) and a public network (e.g. the internet)

checking whether incoming or outgoing data meets a given set of criteria

if the data fails the criteria, the firewall will block the 'traffic' and give the user (or network manager) a warning that there may be a security issue

logging all incoming and outgoing 'traffic' to allow later interrogation by the user (or network manager)

criteria can be set to prevent access to certain undesirable sites; the firewall can keep a list of all undesirable IP addresses

helping to prevent viruses or hackers entering the user's computer (or internal network)

warning the user if some software on their system is trying to access an external data source (e.g. automatic software upgrade); the user is given the option of allowing it to go ahead or requesting that such access is denied.



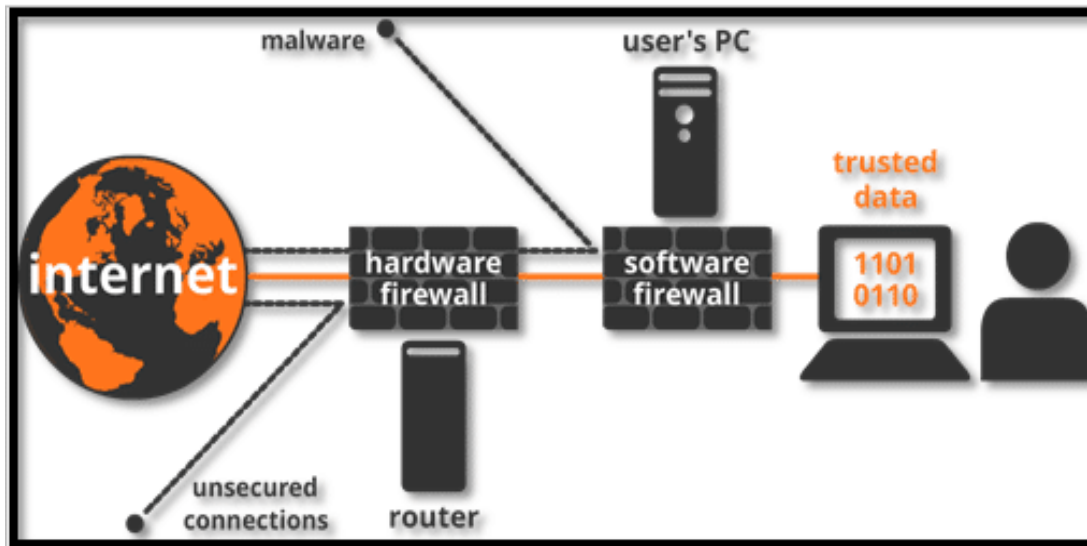
5.3 Cyber security

Keeping data safe from security threats

Firewalls

The firewall can be a hardware interface which is located somewhere between the computer (or internal network external link) and the internet connection. It is often referred to in this case as a **GATEWAY**.

Alternatively, the firewall can be software installed on a computer; in some cases, this is part of the operating system.

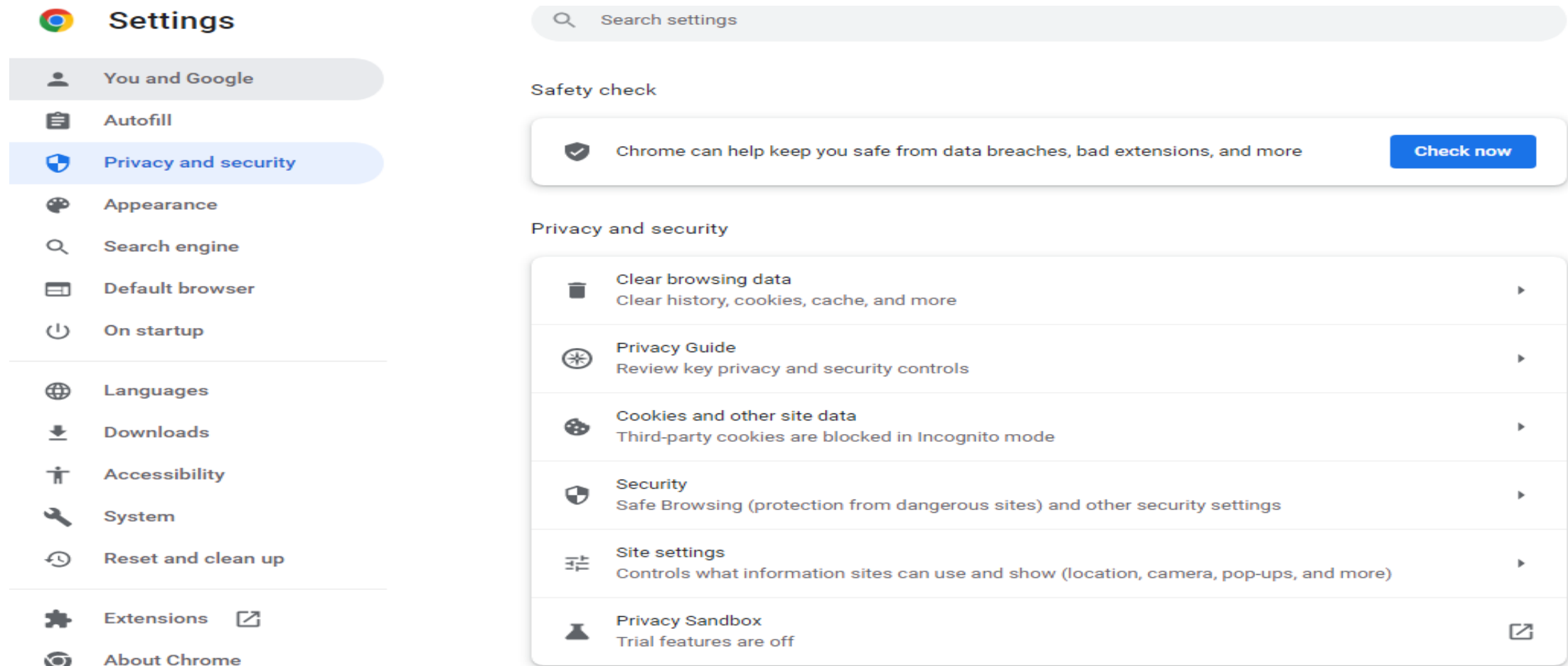


3com FIREWALL PCI CARD hardware ...

5.3 Cyber security

Keeping data safe from security threats

Privacy settings



The screenshot displays the Google Chrome Settings application. On the left, a sidebar lists various settings categories: 'You and Google', 'Autofill', 'Privacy and security' (which is highlighted in blue), 'Appearance', 'Search engine', 'Default browser', 'On startup', 'Languages', 'Downloads', 'Accessibility', 'System', 'Reset and clean up', 'Extensions', and 'About Chrome'. The main content area on the right is titled 'Settings' and features a search bar labeled 'Search settings'. Below the search bar is a 'Safety check' section with a shield icon and the text 'Chrome can help keep you safe from data breaches, bad extensions, and more', accompanied by a 'Check now' button. The 'Privacy and security' section follows, listing several options: 'Clear browsing data' (with a trash icon), 'Privacy Guide' (with a star icon), 'Cookies and other site data' (with a cookie icon), 'Security' (with a shield icon), 'Site settings' (with a list icon), and 'Privacy Sandbox' (with a person icon). Each option includes a brief description and a right-pointing arrow to indicate further settings are available.

Settings

Search settings

Safety check

Chrome can help keep you safe from data breaches, bad extensions, and more [Check now](#)

Privacy and security

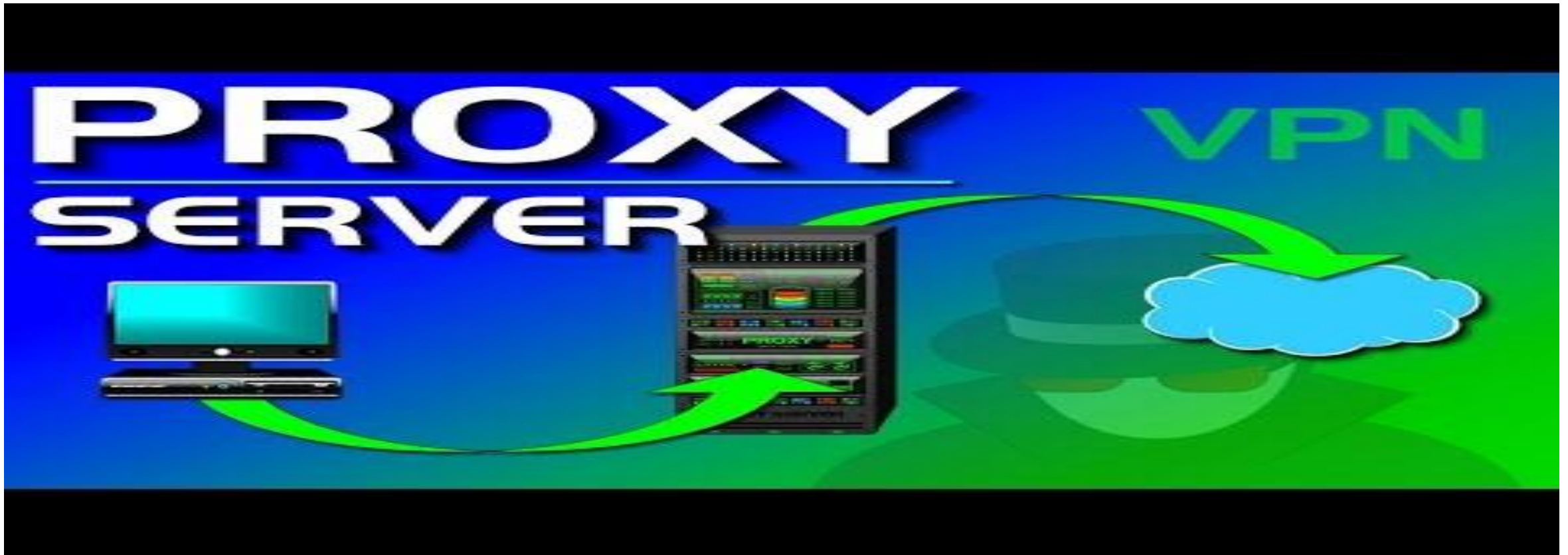
- Clear browsing data**
Clear history, cookies, cache, and more
- Privacy Guide**
Review key privacy and security controls
- Cookies and other site data**
Third-party cookies are blocked in Incognito mode
- Security**
Safe Browsing (protection from dangerous sites) and other security settings
- Site settings**
Controls what information sites can use and show (location, camera, pop-ups, and more)
- Privacy Sandbox**
Trial features are off

5.3 Cyber security

Keeping data safe from security threats

Proxy-Servers

<https://www.youtube.com/watch?v=5cPlukqXe5w>

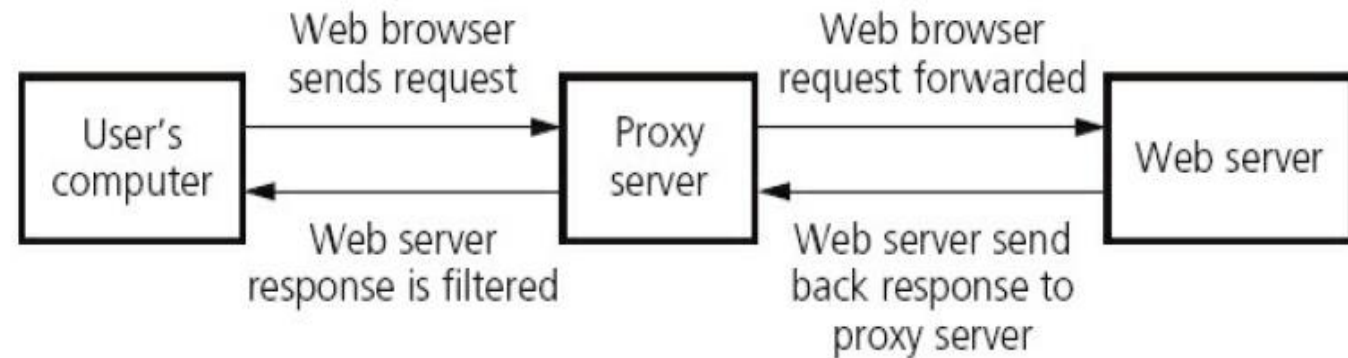


5.3 Cyber security

Keeping data safe from security threats

Proxy-Servers

Act as an intermediary between the user and a web server:



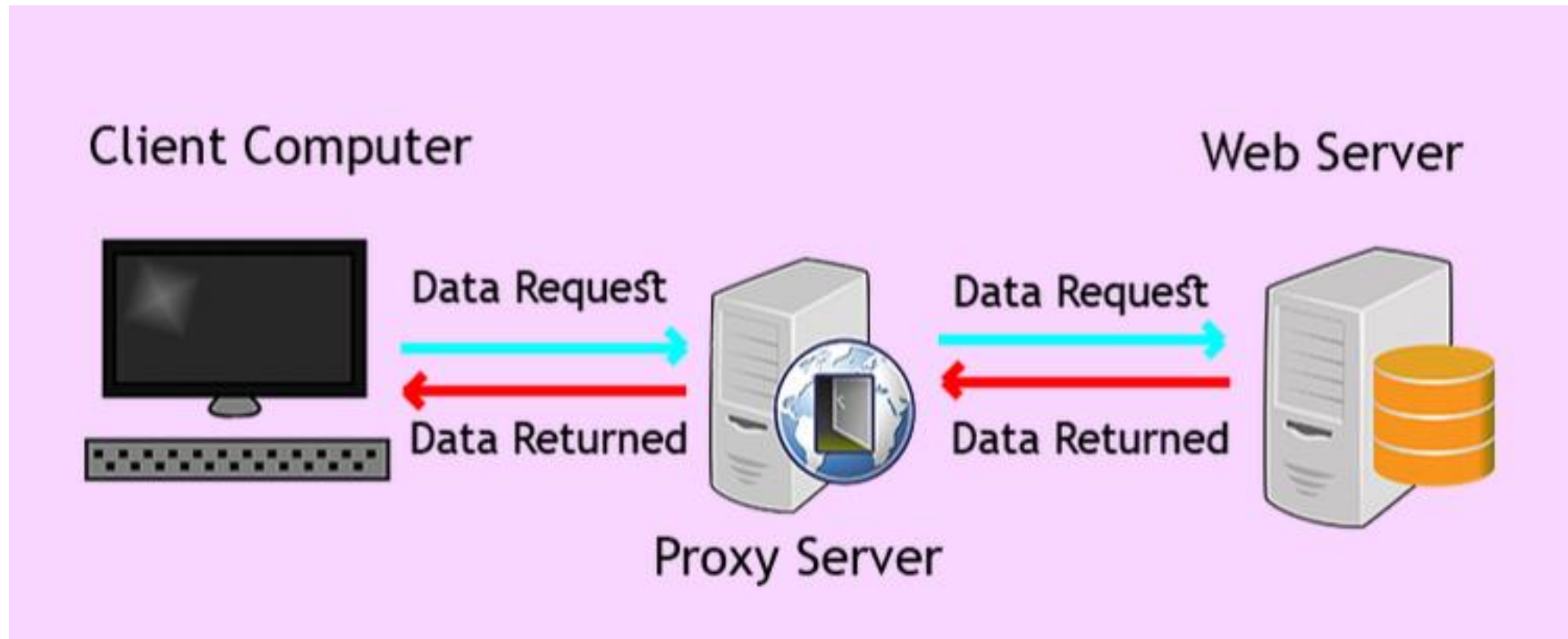
Functions of proxy servers include:

Allowing the internet 'traffic' to be filtered; they can block access to a website if necessary (similar type or reaction as a firewall) by using the feature known as a **CACHE**, they can speed up access to information from a website; when the website is first visited, the home page is stored on the proxy server; when the user next visits the website, it now goes through the proxy server cache instead, giving much faster access keeping the user's IP address secret – this clearly improves security acting as a firewall.

5.3 Cyber security

Keeping data safe from security threats

Proxy-Servers:

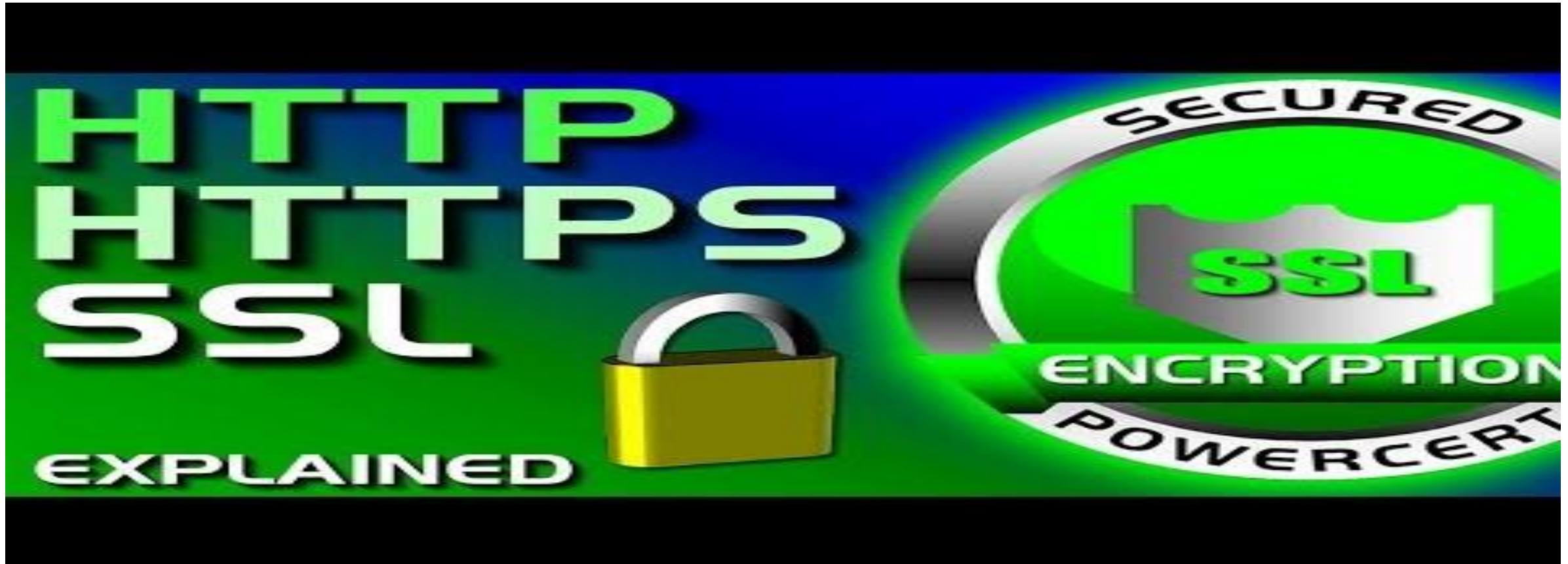


5.3 Cyber security

Keeping data safe from security threats

Secure Socket Layer (SSL) Security Protocol

<https://www.youtube.com/watch?v=hExRDVZHhig>



5.3 Cyber security

Keeping data safe from security threats

Secure Socket Layer (SSL) Security Protocol

<https://youtu.be/J7C9AHvfrrQ>



5.3 Cyber security

Keeping data safe from security threats

Secure Socket Layer (SSL) Security Protocol

https://youtu.be/dsuVPxuU_hc



5.3 Cyber security

Keeping data safe from security threats

Secure Socket Layer (SSL) Security Protocol

SECURE SOCKETS LAYER (SSL): is a type of protocol (a set of rules used by computers to communicate with each other across a network). This allows data to be sent and received securely over the internet.

When a user logs onto a website, SSL encrypts the data – only the user's computer and the web server are able to make sense of what is being transmitted. A user will know if SSL is being applied when they see https or the small padlock  in the status bar at the top of the screen. So what happens when a user wants to access a secure website and receive and send data to it?

5.3 Cyber security

Keeping data safe from security threats

Secure Socket Layer (SSL) Security Protocol

USES OF SSL:

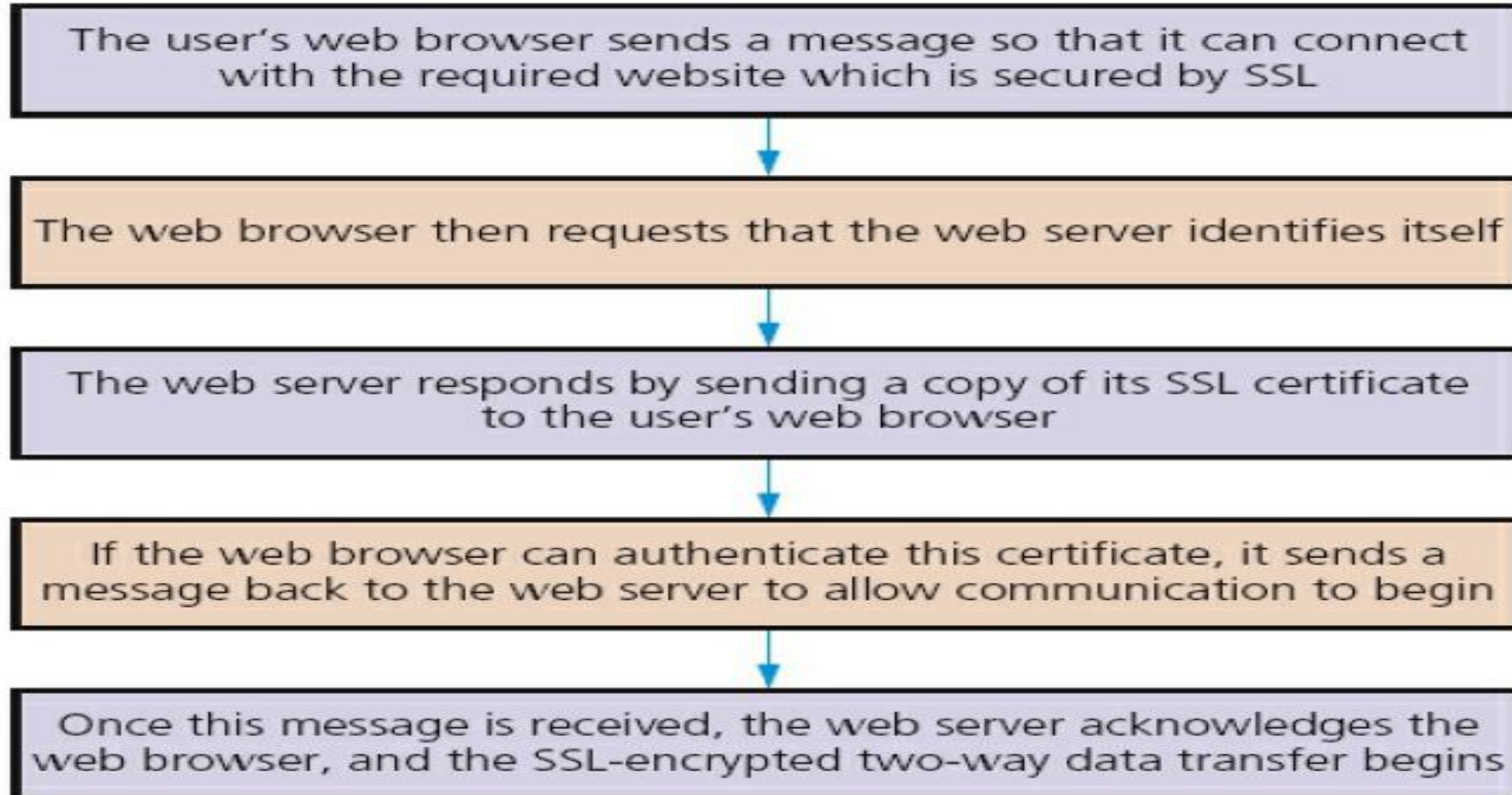
SSL has many practical applications in the real world as security is proving to be These increasingly important, uses include:

- Online shopping / payment systems
- Email
- Cloud based storage e.g. Google Drive
- Intranet / Extranet
- VPN
- VOIP (Video chat)
- Social media
- Online gaming
- Instant messaging

5.3 Cyber security

Keeping data safe from security threats

Security Protocol



IGCSE Exam Question 1

- (a) Viruses, pharming and phishing are all examples of potential Internet security issues.

Explain what is meant by each of these **three** terms.

Virus

.....

.....

.....

Pharming

.....

.....

.....

Phishing

.....

.....

.....

[6]

IGCSE Exam Question 2

Six security issues and six descriptions are shown below.

Draw a line to link each security issue to its correct description.

Security issue

Pharming

Phishing

Viruses

Hacking

Spyware

Cookies

Description

illegal access to a computer system without the owner's consent or knowledge

software that gathers information by monitoring key presses on a user's keyboard; the data is sent back to the originator of the software

malicious code installed on the hard drive of a user's computer or on a web server; this code will re-direct the user to a fake website without the user's knowledge

creator of code sends out a legitimate-looking email in the hope of gathering personal and financial information from the recipient; it requires the user to click on the link in the email or attachment

a message given to a web browser by a web server; it is stored in a text file; the message is then sent back to the server each time the browser requests a page from the server

program or code that replicates itself; designed to amend, delete or copy data or files on a user's computer; often causes the computer to crash or run slowly

IGCSE Exam Question 3

(a) Explain what is meant by a denial of service attack.

.....

.....

.....

.....[2]

(b) Name and describe **two other** potential security threats when using the Internet.

Security threat 1

Description

.....

.....

Security threat 2

Description

.....

.....

[4]

IGCSE Exam Question 4

(c) The company is concerned about a distributed denial of service (DDoS) attack.

(i) Describe what is meant by a DDoS attack.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) Suggest **one** security device that can be used to help prevent a DDoS attack.

..... [1]

IGCSE Exam Question 5

3 Thomas has an online business that sells homemade furniture. He has a web server that hosts his website for his business.

(a) Describe the role of a web browser in requesting and displaying the web pages for the website.

.....

.....

.....

.....

.....

..... [3]

IGCSE Exam Question 5

(b) Thomas is worried about a denial of service (DoS) attack on his web server.

Describe what happens in a denial of service attack.

.....

.....

.....

.....

.....

..... [3]

IGCSE Exam Question 6

- 2 In the diagram, seven descriptions are shown on the left and ten computer terms are shown on the right. By drawing arrows, connect each description to the correct term – each term may be used once or not at all. [7 marks]

users are allowed to try out software free of charge for a trial period; the trial software is often missing certain features

software that can be downloaded from the internet free of charge; the software is subject to copyright laws

set of principles set out to regulate the use of computers, e.g. intellectual property rights

a protocol used by computers on a network to communicate with each other safely

users have the freedom to run, copy, change or adapt this software; it is based on liberty and not price issues

a person takes another person's idea or work and then claims it as their own personal work

attempt at preventing users from accessing an internet server by sending out thousands of requests

free software

cookies

computer ethics

shareware

firewall

plagiarism

encryption

freeware

denial of service attack

secure sockets layer (SSL)

IGCSE Exam Question 7

A company has a number of offices around the world.

- (a) Data is transmitted between the offices over the Internet. In order to keep the data safe the company is using Secure Socket Layer (SSL) protocol and a firewall at each office.

Explain how SSL protocol and a firewall will keep the company's data safe.

SSL protocol

.....

.....

Firewall

.....

.....

[4]

IGCSE Exam Question 7

- (b)** A company stores personal details of its customers on a computer system behind a firewall.

Explain, with reasons, what else the company should do to keep this data safe.

IGCSE Exam Question 8

Six statements about firewalls are shown.

Tick (✓) to show whether each statement is **true** or **false**.

Statement	true (✓)	false (✓)
Firewalls can monitor incoming and outgoing traffic.		
Firewalls operate by checking traffic against a set of rules.		
Firewalls cannot block access to a certain website.		
Firewalls can be software and hardware.		
Firewalls can act as intermediary servers.		
Firewalls can block unauthorised traffic.		

[6]

IGCSE Exam Question 9

- (a) Data is valuable. It needs to be kept secure and it can easily be damaged.

Give **three** different ways that data can be accidentally damaged.

1

.....

2

.....

3

.....

[3]

- (b) The Secure Socket Layer (SSL) protocol can be used to securely transmit data in online banking.

State **three** other different applications that use SSL.

Application 1

Application 2

Application 3

[3]

IGCSE Exam Question 9

(c) Online banking is increasing in popularity.

Online banking can be a risk as it can raise a number of security issues. SSL can be used as a security method to make online banking safer.

Identify and describe **three** other security methods that could be used to make online banking safer.

Security method 1

.....

.....

.....

Security method 2

.....

.....

.....

Security method 3

.....

.....

.....

[6]

IGCSE Exam Question 10

4 Phishing and pharming are two examples of online security threats to a computer system.

(a) Explain what is meant by phishing and pharming.

Phishing

.....

.....

.....

Pharming

.....

.....

.....

[4]

IGCSE Exam Question 10

(b) Identify **two** other online security threats to a computer system.

Security threat 1

Security threat 2 [2]

(c) Give **two** security measures that can help to protect a computer system from online security threats.

Security measure 1

Security measure 2 [2]

IGCSE Exam Question 11

6

- 7 Adele chooses to set a biometric password for her mobile device, instead of a personal identification number (PIN).

(a) State what is meant by a biometric password.

.....
..... [1]

(b) Give **two** reasons why a biometric password is more secure than a PIN.

Reason 1
.....
Reason 2
..... [2]

IGCSE Exam Question 11

(c) Adele has a software-based firewall installed on her mobile device.

The firewall gathers data about the traffic coming into and going out of her mobile device.

Explain how the firewall uses the gathered data to keep the mobile device more secure.

.....

.....

.....

.....

.....

..... [3]

(d) Adele also encrypts the data on her mobile device to keep it more secure.

State how encryption will keep the data more secure.

.....

..... [1]

IGCSE Exam Question 12

(a) The table contains **five** terms or definitions that relate to the Internet.

Complete the table by writing each missing term or definition.

Term	Definition
browser	
.....	this is the company that provides a user with a connection to the Internet
.....	this is a protocol that is used to send data for web pages across the Internet
Uniform Resource Locator (URL)	
cookie	

[5]

IGCSE Exam Question 12

(b) Joelle uses a firewall to keep her data safe when she uses the Internet.

Tick (✓) to show which statement about firewalls is true.

Firewalls can only be hardware-based

Firewalls can only be software-based

Firewalls can be hardware-based or software-based

Tick (✓)

☐☐☐

[1]

(c) Joelle's parent also uses the firewall to limit the websites that Joelle can access.

Explain how the firewall is used to limit the websites that Joelle can access.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

IGCSE Exam Question 13

4 Freddie types a Uniform Resource Locator (URL) into the address bar of his browser.

(a) A URL is a text-based version of a website address. It is made up of several different parts.

State the names of **three** different parts of a URL.

Part 1

Part 2

Part 3 [3]

(b) Explain how the URL is converted into an Internet Protocol (IP) address for the website.

.....
.....
.....
..... [2]

(c) A protocol is used to securely transmit the data for the website to the browser.

(i) State the name of this protocol.

..... [1]

(ii) Identify how this protocol changes the data to transmit the data securely.

..... [1]

IGCSE Exam Answer 1

(a) Question 6 (a) asks you to explain what is meant by three security terms. The question is worth 6 marks but only two marks can be given for each term.

Answers

Virus

Any two from

- Program / software that replicates / copies itself
- Can delete or alter files / data stored on a computer
- Can make the computer "crash" / run slow

Pharming

Any two from

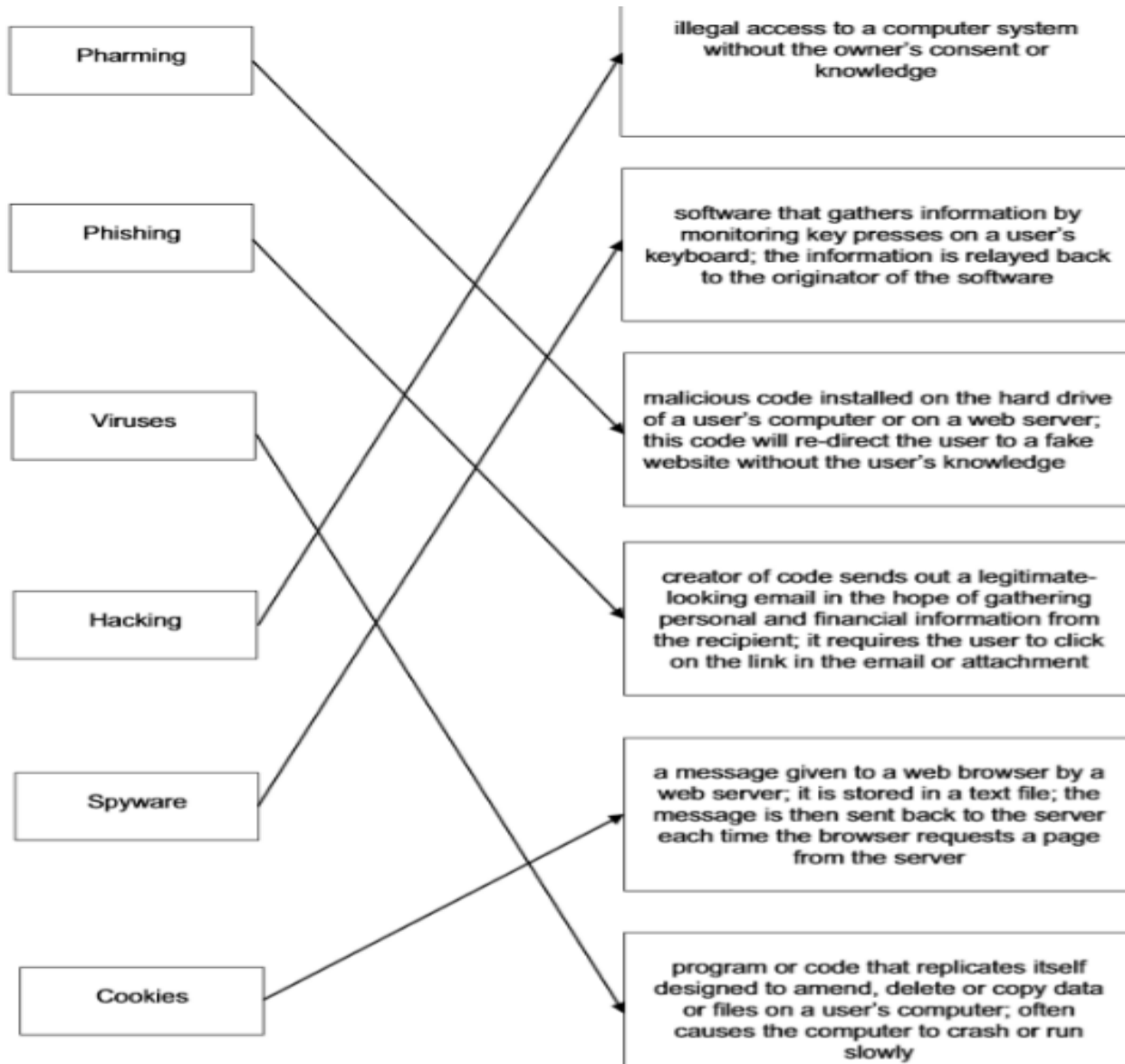
- Malicious code / software installed on a user's hard drive / actual web server
- This code redirects user to a fake website (without their knowledge)
- To obtain personal / financial information / data

Phishing

Any two from

- Legitimate-looking emails sent to a user
- As soon as recipient opens/clicks on the link in the email / attachment
- ...The user is directed to a fake website (without their knowledge)
- To obtain personal / financial information / data

IGCSE Exam Answer 2



IGCSE Exam Answer 3

(a)

Question 9 (a) Asks you to explain what a denial of service attack is. The question is worth 2 marks so you will need to make at least 2 points.

The following points are worth 1 mark each (Max 2)

Answer

- A large number of requests are sent to the network / server all at once
- Designed to flood a network / server with useless traffic / requests
- The network / server will come to a halt / stop trying to deal with all the traffic / requests
- Prevents users from gaining access to a website / server

(b)

Part (b) of question 9 asks you to give and explain two other threats when using the internet. You must remember to explain the threats that you give as there is 1 mark available for the name and 1 mark for the description.

Answers

Viruses

- Software that replicates
- Causes loss / corruption of data // computer may "crash" / run slow

IGCSE Exam Answer 3

Hacking / Cracking

- Illegal / unauthorized access to a system / data

Phishing

- A link/ attachment sends user to a fake website (where personal data may be obtained)

Pharming

- Malicious code installed on user's hard drive / computer
- User is redirected to a fake website (where personal data may be obtained)

Spyware / Key logger

- Send / relay key strokes to a third party

IGCSE Exam Answer 4

2(c)(i)	Any four from: • multiple computers are used as bots • designed to deny people access to a website • a large number / numerous requests are sent (to a server) ... • ... all at the same time • the server is unable to respond / struggles to respond to all the requests • the server fails / times out as a result.	4
2(c)(ii)	firewall OR proxy server	1

IGCSE Exam Answer 5

Question	Answer	Marks
3(a)	Any three from: – Sends request to webserver – Receives web pages back from webserver – Converts HTML to display web page – Manages protocols	3
3(b)	Any three from: – Many requests are sent from a computer – Requests are sent to the <u>webserver</u> – The webserver becomes flooded with traffic – The webserver cannot handle the requests / fails – The website can no longer be accessed – Attack maybe distributed	3

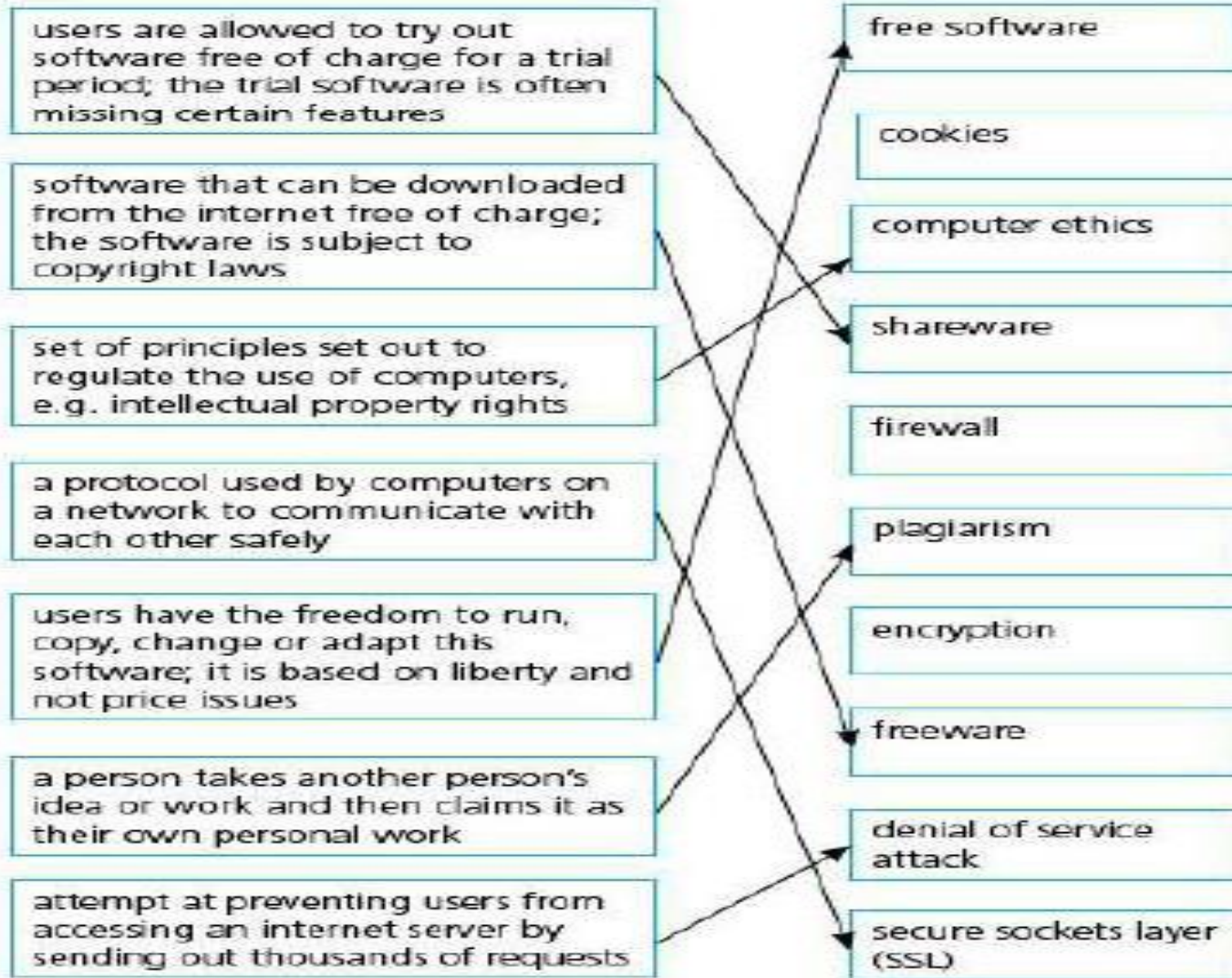
IGCSE Exam Answer 5

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Question	Answer	Marks
4(a)(i)	1 mark for security method, 2 marks for description Anti-virus (software) // Anti-malware (software) • Scans the computer system (for viruses) • Has a record of known viruses • Removes/quarantines any viruses that are found • Checks data before it is downloaded • ... and stops download if virus found/warns user may contain virus Firewall // Proxy server • Monitors traffic coming into and out of the computer system • Checks that the traffic meets any criteria/rules set • Blocks any traffic that does not meet the criteria/rules set // set blacklist/whitelist	3

IGCSE Exam Answer 6

2



IGCSE Exam Answer 7

(a) Question 8 (a) asks for a description of how SSL and Firewalls can help to keep a companies data safe. There are 4 marks available here. 2 marks can be given for SSL and 2 marks can be given for Firewall points.

Answers

SSL Protocol

Two points from

- Uses encryption
- Encryption is asymmetric / symmetric . both
- makes use of (public and private) keys
- Data is meaningless (Without decryption key)

Firewall

Two points from

- Helps prevent unauthorized access // helps prevent hacking
- Checks that data meets criteria
- Acts as a filter for incoming and outgoing data // blocks any unacceptable data

IGCSE Exam Answer 7

(b)

Part (b) requires you to identify methods other than a firewall that a company can use to keep their data safe:

Answers

Question	Answer	Marks
8(b)	Six from: Encrypt the data so it cannot be understood by those not entitled to view it Password protected / biometrics to help prevent unauthorised access Virus checking software helps prevent data corruption or deletion ... identifies / removes a virus in the system ... <u>scans</u> a system for viruses Spyware checking software helps prevent data being stolen/copied/logged ... <u>scans</u> a system for spyware Drop-down input methods / selectable features to reduce risk of spyware / keylogging Physical method e.g. locked doors / CCTV timeout / auto log off ... to help prevent unauthorised access Network / company policies // training employees ... to educate users how to be vigilant Access rights allows users access to data that they have permission to view ... prevents users from accessing data that they do not have permission to view	6

IGCSE Exam Answer 8

Question 7 tests your understanding of firewalls by asking you to identify whether 6 statements about them are true or false.

1 mark is awarded for each correct tick.

Answers

7 Six statements about firewalls are shown.

Tick (✓) to show whether each statement is **true** or **false**.

Statement	true (✓)	false (✓)
Firewalls can monitor incoming and outgoing traffic.	✓	
Firewalls operate by checking traffic against a set of rules.	✓	
Firewalls cannot block access to a certain website.		✓
Firewalls can be software and hardware.	✓	
Firewalls can act as intermediary servers.		✓
Firewalls can block unauthorised traffic.	✓	

[6]

IGCSE Exam Answer 9

(a) Question 8 (a) Asks you to list three different ways that data could accidentally be damaged.

Each of the following points would be worth 1 mark (max 3)

Answers

- Human error e.g deleting / overwriting
- Physical Damage
- Power failure / surge
- Hardware failure
- Software crashing

(b) Question 8 (b) asks you to list 3 other applications that make use of SSL.

Each of the following points would be worth 1 mark (max 3)

Answers

- Online shopping // Online payment systems // online booking
- Email
- Cloud based storage
- Intranet / extranet
- VPN
- VoIP // voice conferencing
- Instant messaging // social networking // online gaming

IGCSE Exam Answer 9

(c) Question (c) asks you to identify and describe three other ways that online banking is made safer. You must remember to explain each point if you wish to gain full marks. 1 mark is awarded for the method and 1 mark is awarded for the description.

Answers

Strong password

- To make it difficult to hack an account

Biometric device

- To use data that is difficult to fake as a password

TLS // Encryption

- To make data meaningless if intercepted
- To encrypt data that is exchanged (TLS only)
- More secure than SSL (TLS only)

IGCSE Exam Answer 9

Firewall

- To help prevent unauthorised access to an account
- Blocks any requests that do not meet / match the criteria

Authentication (card reader at home) / Mobile security code / two - step verification

- To add another level of identification

Use of drop down boxes

- So key loggers cannot record the key presses

Proxy server

- To divert attack away from the main system

IGCSE Exam Answer 10

(a) Question 4 (a) Asks for a description of Phishing and Pharming. The whole question is worth 4 marks so 2 points should be given for each term.

Answers

Phishing

- A Legitimate looking email is sent to a user
- The email will encourage the user to click a link / open an attachment
- The link will redirect a user to a legitimate looking web page (To steal personal info)

Pharming

- A Malicious code is installed on a user's hard drive / server
- The code will cause a redirection to a legitimate looking web page (To steal personal info)

IGCSE Exam Answer 10

(b) Part (b) asks you to list 2 other security threats. Each of the following points would be worth 1 mark (Max 2)

Answers

- Hacking
- Cracking
- Virus
- Denial of Service
- Malware
- Spyware

(c) Part (c) asks you to list two methods to protect your online systems. Each of the following points would be worth 1 mark (Max 2)

Answers

- Firewall
- Proxy Server
- Anti-virus
- Anti-malware
- Anti-spyware
- Username and password

IGCSE Exam Answer 11

Question	Answer	Marks
7(a)	Any one from: • Uses biological data • It uses characteristics/features that belong to a human	1
7(b)	Any two from: • A biometric password cannot be guessed • It is very difficult to fake a biometric password • A biometric password cannot be recorded by a keylogger/spyware • A perpetrator cannot shoulder surf to see a biometric password	2

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Question	Answer	Marks
7(c)	Any three from: • The traffic/data is compared to set criteria/rules/whitelist/blacklist • If the traffic/data does/does not meet the criteria/rules/whitelist/blacklist it will be rejected/blocked ... • ... and an alert can be sent to warn Adele • ... this can help prevent hackers/malicious software (gaining access to the data)	3
7(d)	• It makes the data meaningless	1

IGCSE Exam Answer 12

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Question	Answer	Marks												
3(a)	One mark for the correct term or definition <table><tr><th>Term</th><th>Definition</th></tr><tr><td>browser</td><td>Software/application that allows users to view web pages / render HTML</td></tr><tr><td>Internet Service provider // ISP</td><td>this is the company that provides a user with a connection to the Internet</td></tr><tr><td>HTTP // HTTPS</td><td>this is a protocol that is used to send data for web pages across the Internet</td></tr><tr><td>Uniform Resource Locator (URL)</td><td>a text-based version of a web address</td></tr><tr><td>cookie</td><td>a text file (stored by web browser) that contains data about a user's browsing habits/details/preferences</td></tr></table>	Term	Definition	browser	Software/application that allows users to view web pages / render HTML	Internet Service provider // ISP	this is the company that provides a user with a connection to the Internet	HTTP // HTTPS	this is a protocol that is used to send data for web pages across the Internet	Uniform Resource Locator (URL)	a text-based version of a web address	cookie	a text file (stored by web browser) that contains data about a user's browsing habits/details/preferences	5
Term	Definition													
browser	Software/application that allows users to view web pages / render HTML													
Internet Service provider // ISP	this is the company that provides a user with a connection to the Internet													
HTTP // HTTPS	this is a protocol that is used to send data for web pages across the Internet													
Uniform Resource Locator (URL)	a text-based version of a web address													
cookie	a text file (stored by web browser) that contains data about a user's browsing habits/details/preferences													
3(b)	• Hardware or software based	1												
3(c)	Four from: • (The parent can) set criteria for the websites she is allowed to visit • ... such as a whitelist/blacklist of websites • The firewall will examine the data/traffic incoming and outgoing from her computer • If data is sent from a website that is not allowed, it will be blocked	4												

IGCSE Exam Answer 13

Question	Answer	Marks
4(a)	Three from: • Protocol • Domain name / Web server name • Filename / web page name / folder name	3

PUBLISHED

Question	Answer	Marks
4(b)	• It is sent to a DNS ... • which looks up the corresponding/matching IP address	2
4(c)(i)	• HTTPS	1
4(c)(ii)	Any one from: • It encrypts it • It applies encryption algorithm • It applies an encryption key	1